

F24-119-D-ArenaX

Project Team

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Chapter 1

Introduction

The following Software Requirement Specification (SRS) document outlines the design, architecture, functional and non-functional requirements of our application i.e. **ArenaX**, a cross-platform social-professional networking app designed for the gaming community. This document ensures the clear understanding of the project vision, structure, features, constraints and project goals and so is intended for stakeholders including FYP evaluation committee, project development team and app users.

1.1 Problem Statement

In today's digital world, we consume content and share selective aspects of our lives tailored to the niche of each networking platform; however when it comes to gaming, there is an absence of a centralized networking platform for this billion-dollar industry. Currently, a global audience of 3.2 billion is dispersed and segmented across multiple social media platforms including Facebook groups, Reddit communities and servers on Discord. This user fragmentation causes multiple networking issues for the gamers including absence of gaming portfolios subsequent to having social profiles across social media apps, unavailability of focused networking tools for maximizing gamer interaction based on shared interests, lack of a platform for discovery and participation in e-sports and an almost negligible online presence of e-sports as a corporate industry around the world.

Our app ArenaX aims to be the centralized networking platform serving the gaming industry both as a social and a professional/corporate platform. With our user-friendly features we want to ensure increased interaction and collaboration in a centralized environment, providing recognition to casual gamers and e-athletes and serving as a professional platform for the esports industry.

1.2 Scope

The proposed project aims at developing a cross-platform social networking platform. Our project's scope includes the development of both a desktop application as well as a mobile application to cater to cross-platform interactions. Our team will start working on the design of our app's user interface and gathering datasets on which the recommender model will be trained. In subsequent iterations the team will focus on developing the user profile module first followed by other modules. It is to be noted that the desktop app will consist of the user profile module only. The player profile module will consist of user profiles where user can list the games they play and our app will be responsible for tracking the screen time of each game listed on the user's profile. It would also enable user to share gameplay and content. This module will be followed by developing the recommender model in addition to implementing the user feed and connection system module simultaneously. The recommender model will be developed from scratch using gaming datasets and would be trained to function based on shared gaming niche and interests in addition to incorporating Geo-location of users. The connection system is a networking module that would empower the app's users to connect with each other based on shared interests. Lastly the modules related to esports organizational management would be developed and the recommender model designed earlier would be incorporated in the features where necessary. This module will provide features including tournament/event organization and listing, team and staff recruitment and sponsored posts. A detailed explanation of each module is listed below.

1.3 Modules

Following is the modular division of our app's features.

- Desktop App
 1. User profile
- Mobile App
 1. User profile and analytics
 2. Networking system
 3. Power play
 4. Targeted ads and sponsored posts

1.3.1 Module 1: User Profile (Desktop App)

The desktop application will consist of the user profile module only. This module mainly focuses on screen-time tracking for each game the user plays and creating various stats on the basis of that data in addition to simple profile management features. It will also enable user to post gameplay and achievements on their profile. The main features of this module include:

1. Personal profile management
2. Playtime analytics dashboard including total playtime across all games, total play-time per game, average playtime, peak playtime, favorite game, favorite genre
3. Post gameplay and achievements
4. Achievement/ranking system based on total playtime across all games played.

1.3.2 Module 2: User Profile (Mobile App)

This module of the mobile application is similar to that of the desktop application and provides the same features as mentioned below.

1. Personal profile management
2. Playtime analytics dashboard including total playtime across all games, total play-time per game, average playtime, peak playtime, favorite game, favorite genre
3. Post gameplay and achievements
4. Achievement/ranking system based on total playtime across all games played.

1.3.3 Module 3: Networking System (Mobile App)

This module focuses on the development of features related to user engagement and interaction with peers across the app and leverages a recommender system.

1. Interest-based user feed
2. Synergy (Connection system)
3. Threads (Chats)
** Please note that feature 1 and 2 will leverage a recommender system that will cater to user's gaming interests and Geo-location*

1.3.4 Module 4: Power Play (Mobile App)

This module covers the professional niche of our app and offers a number of features for organizational management and networking. The main features of this module are listed below.

1. Organization profile management
2. Battlegrounds (Tournaments/ events listings)
3. TalentXchange (Staff recruitment listings)
** Please note that feature 2 and 3 will leverage a recommender system that will cater to user's gaming interests and Geo-location*

1.3.5 Module 5: Targeted Ads (Mobile App)

This module will enable the user to enhance the reach of their posts/ads amongst an audience that meets the post's niche. It's features are mentioned below

1. Sponsored posts
2. Targeted advertisements
** Please note that feature 1 and 2 will leverage a recommender system that will cater to user's gaming interests and Geo-location*

1.4 User Classes and Characteristics

Following is the list of the user classes that will use our app.

1. Causal gamer
2. Content creator
3. Esports athlete
4. Esports organization
5. Game company
6. Advertiser

User class	Description
Casual gamer	gamer who play games just for fun and entertainment and will use ArenaX primarily for maintaining player portfolio, networking with gamers having similar interests and discovering content relevant to the user's interests. The users will access the recommender system for connections and content feed. Majority of these users will access the app from mobile phones while some may use the desktop app to maintain their player profile
Content creator	gamer who streams, records or shares content on ArenaX to build an audience by engaging app users with their gameplay content. These users will mostly utilize the player profile feature, content feed and connection system feature to build their audience on the app. They may use the mobile or desktop app to access these features.
Esports athlete	a professional gamers who play competitive games and will use ArenaX to build their portfolio, discover and participate in game events and tournaments, connect with esports teams and organizations. These users can discover and avail opportunities in the esports industry through listings in the Job market. These users may use the desktop or mobile app for building their gaming portfolios but the other features mentioned above are only available on the mobile application
Esports organization	Esports company or organization will use ArenaX to build their company portfolio, recruit players and staff, manage teams and manage game events and tournaments. The features mentioned above are only available on the mobile app and so the users may access and avail them through our mobile application
Game company	Game companies will use ArenaX to post their game updates, updates about new games, releasing games through ArenaX and marketing through our app. They will use the mobile app to access these features.
Advertiser	Any individual, organization or company can increase the reach or advertise their content using our targeted advertisements feature. This feature is also platform-specific and is only available on the mobile app.

Example: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-cases

The use case diagram of ArenaX is depicted below.

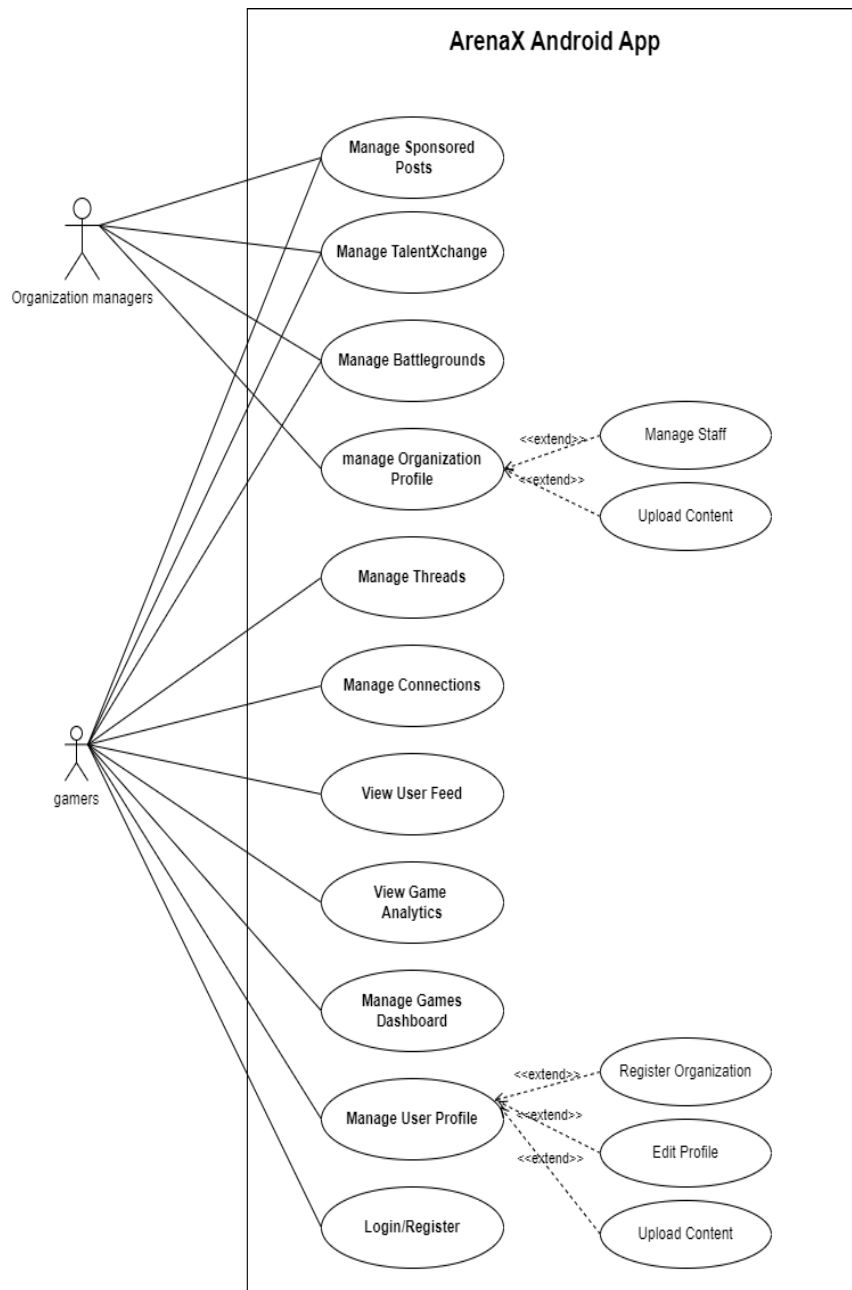


Figure 2.1: ArenaX Use case Diagram

Following are the detailed use cases of the project.

1. UC 01 - Register User:

Use Case ID	UC 01
Use Case Name	Register user
Scope	ArenaX app
Primary Actor	• Casual gamers • Content creators • Esports athletes • Company managers
Description	All user classes who want to access the app features need to register themselves on the app
Stakeholders and Interests	All user classes who want to use the app and avail all the features provided by the application.
Pre-Conditions	• The user has access to ArenaX mobile or desktop app. • The user has an active email or google account.
Post-Conditions	• User is successfully registered and can login. • User's profile is created on the app. • User data is stored in database. • User can now access all of the app's features.
Success Guarantee	User data and credentials will be stored and authenticated in the database.
Alternatives	User chooses to register on the app using their Google account instead of email.

Extensions	• If the user enters an already registered email address or google account, an error message will prompt to try another email. • If the user enters an already used gamer-tag, an error message will prompt to enter a unique gamer-tag. • In case of no internet access, the user will be prompted to check for valid internet connection.
Exceptions	User enters incomplete or invalid credentials and personal information.
Special Requirements	Validate email through code or confirmation link
Technology and Data Variation	Android and Desktop app must have the same registration process.
Frequency of Occurrence	High, as new users must Sign up to continue.
Miscellaneous	None

Step	User Action	System Response
1	User opens ArenaX app	Splash Screen appears and user is directed to login page.
2	User Clicks on the " Don't have an account? Sign Up " link	User is redirected to the registration screen
3	User enters email, password and confirms password	Data is saved temporarily
4	User clicks on the next button	User is directed to the personal information page
5	User enters information including full name, date of birth, gender and unique gamer-tag	User's information is temporarily saved
6	User clicks on the Sign Up button	The system authenticates user data and credentials and sends a confirmation link on user's email.
7	User confirms the email	The system verifies the confirmation, creates and activates user's account
8	User can now login and access app features	The system automatically logs in the user into his account.

2. UC 02 - User Login:

Use Case ID	UC 02
Use Case Name	User Login
Scope	ArenaX app
Primary Actor	• Casual gamers • Content creators • Esports athletes • Company managers
Description	All registered user classes who want to access the app features need to log in.
Stakeholders and Interests	All user classes who want to use the app.
Pre-Conditions	• The user is registered on the app. • The user has active internet connection.
Post-Conditions	• User is successfully logged in. • User can now access all in-app features.
Success Guarantee	User credentials are authenticated against the data stored in the database.
Alternatives	• User redirects to the sign-up page. • User chooses to recover the password instead of logging in.
Extensions	In case of invalid credentials being entered, an appropriate error message is displayed.
Exceptions	User account does not exist.
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Highly frequent, as users must log in to use the app.

Miscellaneous	None
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Step	User Action	System Response
1	User opens ArenaX app	Splash Screen appears and user is directed to the login page.
2	User enters their credentials, including email and password	The system validates the user's credentials.
3	User is logged in and redirected to the home page	The system displays the user's content feed on the home page.

3. UC 03 - Manage User Profile:

Use Case ID	UC 03
Use Case Name	Manage user profile
Scope	ArenaX app
Primary Actor	• Casual gamers • Content creators • Esports athletes • Company managers
Description	User can manage their profiles including updating account information, registering organization, and uploading content.
Stakeholders and Interests	All user classes who wish to build and maintain their profiles within the app.
Pre-Conditions	• The user is logged in.
Post-Conditions	• User's profile is successfully updated. • Successful registration of the organization. • Uploaded content is stored and accessible on the user's profile.
Success Guarantee	User data is stored in the database and reflected in the user's account.
Alternatives	None

Extensions	• If the user enters invalid data during profile editing, an appropriate error message is displayed. • If the organization already exists during registration, the user is prompted to choose a different name. • In case of a failed upload, the user is notified.
Exceptions	None
Special Requirements	Data validation and confirmation boxes for profile changes.
Technology and Data Variation	None
Frequency of Occurrence	Frequent, as users often update profiles and upload content.
Miscellaneous	None

Step	User Action	System Response
1	User navigates to the profile page	User data is loaded.
2	User clicks on the " edit profile " button	The system redirects the user to the edit profile page.
3	User updates account information	User data is validated, and the system asks for confirmation.
4	User confirms the change	The system updates user data, and the changes are reflected in the account.
5	User clicks on the upload button in the navigation bar and chooses a file	The system prepares the file for upload and asks for confirmation.
6	User confirms the file upload	The file is uploaded and reflected in the user's account.
7	User clicks on the register organization tab	The system redirects the user to the organization registration form.
8	User enters data and clicks on register	The data is validated, and the organization's page is created and listed on the app.

4. UC 04 - Manage Games Dashboard:

Use Case ID	UC 04
Use Case Name	Manage Games Dashboard
Scope	ArenaX app
Primary Actor	• Casual gamers • Content creators • Esports athletes
Description	User can manage their game dashboard by adding games, and the system will automatically create analytics for each game based on screen time.
Stakeholders and Interests	All gamers who wish to create their game portfolio through screen-time analytics for each game.
Pre-Conditions	• The user is logged in. • User has listed a game on their dashboard.
Post-Conditions	• The game is successfully added to the user's dashboard. • The system generates analytics based on the screen time of the game when it is played on the device.
Success Guarantee	The game dashboard is updated, and a tab is created for the listed game to display statistics for that game.
Alternatives	• User tries to add a game already listed on the dashboard. • User removes a game from the dashboard.
Extensions	• If the game is not found on the user's device, an appropriate error message is displayed. • The user enters an invalid name for a game.
Exceptions	None

Special Requirements	• Game statistics are updated in real time. • The system validates game titles for legitimacy.
Technology and Data Variation	None
Frequency of Occurrence	Highly frequent
Miscellaneous	None

Step	User Action	System Response
1	User navigates to the Game dashboard page	User's current games list is displayed.
2	User clicks on the "Add game" button	The system displays a search bar to select the game.
3	User selects the game	The game is added to the list on the dashboard.
4	User plays the game	The system calculates screen time for that game and displays various metrics based on the screen time.

5. UC 05 - View Game Analytics:

Use Case ID	UC 05
Use Case Name	View Game Analytics
Scope	ArenaX app
Primary Actor	• Casual gamers • Content creators • Esports athletes • Company Managers
Description	Users can view detailed user statistics of the games listed on a user's game dashboard.

Stakeholders and Interests	• Users wanting to track their game time and build a gaming portfolio. • Organizations interested in reviewing game-related statistics for potential recruitment.
Pre-Conditions	• The user has added a game to their dashboard.
Post-Conditions	• User can view game statistics including total hours spent across games, total hours in a specific game, peak playtime graph, and average playtime graph.
Success Guarantee	All statistics are accurately displayed.
Alternatives	None
Extensions	None
Exceptions	System fails to retrieve data.
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Highly frequent, as users check their statistics often.
Miscellaneous	None

Step	User Action	System Response
1	User navigates to their profile page	The system displays user data which includes the user's games list.
2	User clicks on a game in the game dashboard list	The system redirects the user to the game analytics page and displays detailed statistics for that game.

6. UC 06 - View User Feed:

Use Case ID	UC 06
Use Case Name	View User Feed
Scope	ArenaX app
Primary Actor	• Casual gamers • Content creators • Esports athletes • Company managers
Description	Users can view a personalized feed that includes posts and game-play videos according to their interests and recent activity.
Stakeholders and Interests	• Users who want to stay updated with gaming content and posts from peers. • Content creators wishing to promote their content and build an audience. • Organizations posting updates and publicizing their ventures. • Game companies posting updates about their games and new arrivals.
Pre-Conditions	• The system has data about the user's interests.
Post-Conditions	• User views a feed containing content of their interest.
Success Guarantee	Relevant content is displayed for each user.
Alternatives	User manually searches for content.
Extensions	None
Exceptions	None
Special Requirements	Real-time updation of the content feed on arrival of new data.
Technology and Data Variation	None
Frequency of Occurrence	Most Frequent, as users interact with content feeds frequently in a day.

Miscellaneous	None
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Step	User Action	System Response
1	User navigates to the home page	The system loads the content feed.
2	User scrolls the feed	The system loads and displays newer content upon user scroll.
3	User interacts with the post by liking, commenting, or sharing the post	The interaction is reflected on the post and, upon sharing, the content is available on the user's profile.

7. UC 07 - Manage Connections:

Use Case ID	UC 07
Use Case Name	Manage Connections
Scope	ArenaX app
Primary Actor	• Casual gamers • Content creators • Esports athletes • Company managers
Description	Users can search for, send requests to, and manage existing connections with other users on the app.
Stakeholders and Interests	• Users wanting to expand their network and connect with gamers with shared interests. • Content creators wishing to promote their content and build an audience. • Organizations looking for collaborators, talent, and audience engagement.
Pre-Conditions	• The user is logged in. • There are other users on the app.

Post-Conditions	• User successfully sends or receives a connection request. • Any update in a user's connections is reflected in the connections list. • The system recommends users based on region and shared interests.
Success Guarantee	A connection request is successfully sent, received, or established between two users.
Alternatives	None
Extensions	User blocks or removes a connection.
Exceptions	None
Special Requirements	None
Technology and Data Variation	The system should leverage a recommender model that uses both shared interests and geo-location of users for recommendations.
Frequency of Occurrence	Frequent, as users are likely to connect with other users.
Miscellaneous	None

Step	User Action	System Response
1	User navigates to the connections page	The system recommends a list of users with shared interests.
2	User selects another user from the list or through the search bar and sends a request	The system sends the request to the receiver and provides a confirmation to the sender.
3	User accepts or declines an incoming request	The system updates the user's connection list accordingly.
4	User removes a connection with a user	The system updates the user's connection list accordingly.

8. UC 08 - Manage Threads:

Use Case ID	UC 08
Use Case Name	Manage Threads
Scope	ArenaX app
Primary Actor	• Casual gamers • Content creators • Esports athletes • Company managers
Description	Users can send and receive chats, contacting other users on the app.
Stakeholders and Interests	• Users wanting to communicate with other users on the app. • Content creators wishing to communicate with their audience. • Organizations wishing to coordinate with their staff.
Pre-Conditions	• The user is logged in. • There are other users on the app.
Post-Conditions	• User successfully sends or receives a message. • Logs are maintained for messages and stored in the database.
Success Guarantee	A message is successfully sent or received by a user.
Alternatives	None
Extensions	User deletes a conversation.
Exceptions	Connectivity issues preventing message delivery.
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Highly frequent, as typically users connect with each other daily.
Miscellaneous	None

Step	User Action	System Response
1	User navigates to the threads page	The system displays a list of current conversations.
2	User chooses from the list or searches for a new user to text	A conversation is created for that user, and the user is navigated to the chat page.
3	User sends a message	The system sends the message to the receiver.
4	User deletes a conversation	The system deletes the conversation and updates the list.

9. UC 09 - Manage Organization Profile:

Use Case ID	UC 09
Use Case Name	Manage Organization Profile
Scope	ArenaX app
Primary Actor	• Company managers
Description	Allows managers to manage their organizational profiles, including managing staff, assigning positions, and posting content related to team updates and achievements.
Stakeholders and Interests	• Esports organizations who want to establish their online presence and manage their staff. • Team members need to view their roles and link with an organization. • Fans interested in organizational updates and content.
Pre-Conditions	• The user must have registered an organization on the app. • The user must have administrative access to the organization.
Post-Conditions	• The organization's profile is updated in case of any changes. • Members are notified in case of any changes to their role. • Content is successfully uploaded on the organization's profile.
Success Guarantee	Changes are reflected on the organization's profile.

Alternatives	None
Extensions	None
Exceptions	None
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Occasionally, as such changes do not occur on a daily basis.
Miscellaneous	None

Step	User Action	System Response
1	User navigates to the organization profile page	The system loads user data.
2	Organization admin selects the option to add a staff member	The system prompts the admin to enter details of the new member.
3	Admin enters data and sends a request to the user to accept their role	The system forwards the request to the targeted user.
4	Admin clicks on the update roles button	The system displays a list of all enrolled members along with their roles.
5	Admin chooses a new role for a member	The system updates the changes made.
6	Admin removes a member from the organization	Changes are reflected, and organizational access for that member is ceased.
7	Admin clicks on the upload button and selects files for upload	The system loads the files and prompts the user to confirm the upload.
8	Admin confirms the upload	The content is uploaded on the organization's profile and is visible on the app.
9	Admin clicks on the edit profile button	The system redirects to a page to edit the organization's information.
10	Admin commits changes and clicks on update	The system validates the changes and updates the organization's profile after successful validation.

10. UC 10 - Manage Battlegrounds:

Use Case ID	UC 10
Use Case Name	Manage Battlegrounds
Scope	ArenaX app
Primary Actor	• Casual gamers • Content creators • Esports athletes • Company managers
Description	Allows organizations to register and organize different game tournaments. It also allows players to view and register in these tournaments.
Stakeholders and Interests	• Esports organizations who want to organize and promote their tournaments. • Players who want to participate in these tournaments. • Team managers who want to register their team in these tournaments.
Pre-Conditions	The user must have an organization profile to register a tournament.
Post-Conditions	• Player can register in a tournament successfully. • An organization can successfully register a tournament.
Success Guarantee	Organization successfully lists a tournament and it can be viewed on the battlegrounds page.
Alternatives	None
Extensions	• Player cancels a registration. • Organization updates tournament details.
Exceptions	None
Special Requirements	None
Technology and Data Variation	None

Frequency of Occurrence	Highly frequent, as tournaments are organized and played on a daily basis.
Miscellaneous	None

Step	User Action	System Response
1	Organization admin navigates to the organization profile page	The system loads relevant data of the page.
2	Admin clicks on the organize tournament button	The system displays a tournament form containing fields for all relevant details.
3	Admin fills the form and submits it	The system stores the form and lists the tournament in the Battlegrounds tab.
4	Player navigates to the Battlegrounds page	The system loads a list of registered tournaments.
5	The player clicks on a tournament to view more details	User is redirected to a page where detailed information about the tournament is displayed.
6	Users can filter the tournaments list using various filters including game, tier, and region	The system updates and displays the list accordingly.
7	Player clicks on the register button	The system navigates the user to a form created by the tournament's management team to register in the tournament.
8	User fills the form and submits it	The system forwards the form to the organization and notifies the user about it.

11. UC 11 - Manage TalentXchange:

Use Case ID	UC 11
Use Case Name	Manage TalentXchange
Scope	ArenaX app
Primary Actor	• Casual gamers • Content creators • Esports athletes • Company managers
Description	Allows organizations to list job openings and opportunities. Users can view and apply for these job listings.

Stakeholders and Interests	• Esports organizations who want to hire players and staff for their organization. • Users who want to secure an opportunity relevant to their skills and experience.
Pre-Conditions	The user must have an organization profile to list a job.
Post-Conditions	• An organization can successfully list a job and manage received applications. • Players can view and apply for jobs.
Success Guarantee	Organization successfully lists a job opportunity and it can be viewed in the TalentXchange section.
Alternatives	None
Extensions	• User withdraws application for a job. • Organization closes a job listing.
Exceptions	None
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Frequently, as jobs are frequently listed and applied for.
Miscellaneous	None

Step	User Action	System Response
1	Organization admin navigates to the organization profile page	The system loads relevant data of the page.
2	Admin clicks on the create job listing option	The system displays a job listing form containing fields for all relevant details.
3	Admin fills the form and submits it	The system stores the form and lists the job in the TalentXchange section.
4	User navigates to the TalentXchange page	The system loads a list of available jobs.

5	The user clicks on a job to view more details	User is redirected to a page where detailed information about the job is displayed.
6	Users can filter the jobs list using various filters including game, job type, and region	The system updates and displays the list accordingly.
7	Player clicks on the apply button	The system navigates the user to a form created by the organization's management team to apply for the job.
8	User fills the form and submits it	The system forwards the form to the organization and notifies the user about it.

12. UC 12 - Manage Sponsored Posts:

Use Case ID	UC 12
Use Case Name	Manage Sponsored Posts
Scope	ArenaX app
Primary Actor	• Casual gamers • Content creators • Esports athletes • Company managers
Description	Enables all user classes to increase their post engagement on the platform through sponsored posts for a targeted audience.
Stakeholders and Interests	• Esports organizations who want to promote their events, tournaments, or recruitment posts for better reach. • Content creators who want to boost their post visibility. • Game companies who want to promote their newly launched games, in-game events, or updates.
Pre-Conditions	The user must have registered the subscription plan for sponsored posts.

Post-Conditions	• A post is visible to newer audiences of relevant interests. • The post's performance can be tracked through analytics.
Success Guarantee	The sponsored post has increased visibility and better reach in the targeted audience.
Alternatives	None
Extensions	None
Exceptions	User's subscription plan has expired and so cannot sponsor a post.
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Frequently, as users regularly create and sponsor posts.
Miscellaneous	None

Step	User Action	System Response
1	The user clicks on the "upload" button in the navigation bar and selects a file for upload	The system loads the file and prompts the user to confirm upload.
2	User clicks on the "Sponsor post" option	The user is redirected to a form to sponsor a post.
3	User fills in the details including target audience and subscription plan and submits the form	The system validates the validity of the subscription plan and ensures increased visibility of the post.

2.2 Functional Requirements

Below listed are the functional requirements of each module of our app.

2.2.1 Module 1: User Profile (Desktop App)

Following are the requirements for module 1:

1. The system shall allow users to register on the app, create and edit their profiles.
2. The system shall allow users to upload content including text, photos and videos on their profile.
3. The system shall allow users to list games on their dashboards.
4. The system shall fetch screen time and generate analytics for the games played by a user.
5. The system shall record and maintain user's game-time analytics.
6. The system shall track and display total hours spent across all games.
7. The system shall rank users based on their total playtime across all games.

2.2.2 Module 2: User Profile (Mobile App)

Following are the requirements for module 2:

1. The system shall allow users to register on the app, create and edit their profiles.
2. The system shall allow users to upload content including text, photos and videos on their profile.
3. The system shall allow users to list games on their dashboards.
4. The system shall fetch screen time and generate analytics for the games played by a user.
5. The system shall record and maintain user's game-time analytics.
6. The system shall allow users to view profiles, posts and stats of other users on the app.
7. The system shall track and display total hours spent across all games.
8. The system shall rank users based on their total playtime across all games.

2.2.3 Module 3: Networking System (Mobile App)

This module includes personalized user feed, Synergy (connection system) and Threads (chats). Following are the requirements for this module.

1. The system shall display a personalized feed for users according to their gaming interests.
2. The system shall update user feed in real-time upon arrival of new posts.
3. The system shall recommend potential connections based on similar gaming interests and region.
4. The system shall allow users to send, accept or reject a synergy request.
5. The system shall allow users to view their synergy list.
6. The system shall allow users to start conversations with their connections.
7. The system shall allow users to send text, photos or videos in a message.
8. The system shall allow users to delete a conversation.
9. The system shall allow real-time messaging between users.

2.2.4 Module 4: Power Play (Mobile App)

This module includes Organization profile management, Battlegrounds (Tournaments/event listings) and TalentXchange (Staff recruitment). Following are the requirements for this module.

1. The system shall allow users to register organization, create and edit organizational profile.
2. The system shall allow an organization's admin to manage staff including adding, removing and roles assignment.
3. The system shall allow an organization's admin to post content including text, photos or videos.
4. The system shall allow organizations to create, list and manage tournaments.
5. The system shall display a personalized list of tournaments and events for each user based on their gaming interests and Geo-location.

6. The system shall allow app users to view and register in tournaments listed on the app.
7. The system shall allow organizations to post job openings.
8. The system shall display a personalized list of job opportunities based on user's interests and region.
9. The system shall allow users to view and apply for these jobs listed on the app.

2.2.5 Module 5: Targeted Ads (Mobile App)

This module includes sponsored posts and advertisements. Following are the requirements for this module.

1. The system shall allow users to create sponsored posts.
2. The system shall allow users to select their target audience and view count while sponsoring posts.
3. The system shall ensure enhanced visibility and reach of sponsored posts.
4. The system shall allow users to view analytics of their sponsored posts.

2.3 Non-Functional Requirements

Following are the non-functional requirements of the system.

2.3.1 Performance

1. The system shall handle up to 1000 users without performance degradation in the initial stage.
2. The response time for user input including posting and profile management must be within 4 seconds.
3. The system shall allow smooth navigation in the app and must happen within 1 seconds of the event.

2.3.2 Availability

1. The system must have 99.99% up-time to ensure availability to users.
2. Features including profile management, game-time tracking and messaging must always be available irrespective of app traffic.

2.3.3 Accessibility

1. The app will only run in presence of stable internet connection.

2.3.4 Scalability

1. The app must be able to handle influx of new users, at least 5000.
2. The analytics system must be able to handle large amounts of fetched data.

2.3.5 Security

1. User data is secured using Firebase security and account credentials are hashed using SHA-256 and so will take a thousands of years of continuous computational effort for a possible breach.

2.3.6 Usability

1. The system shall have a user-friendly interface.
2. The system shall be easy to navigate for all user classes.
3. Users shall be able to access all features easily with minimal effort and learning curve.

2.3.7 Reliability

1. The system must recover without any data loss after a system crash.
2. Playtime tracking and analytics calculations must be accurate and reliable.

2.4 Domain Model

Following is the extensive domain model of ArenaX app.

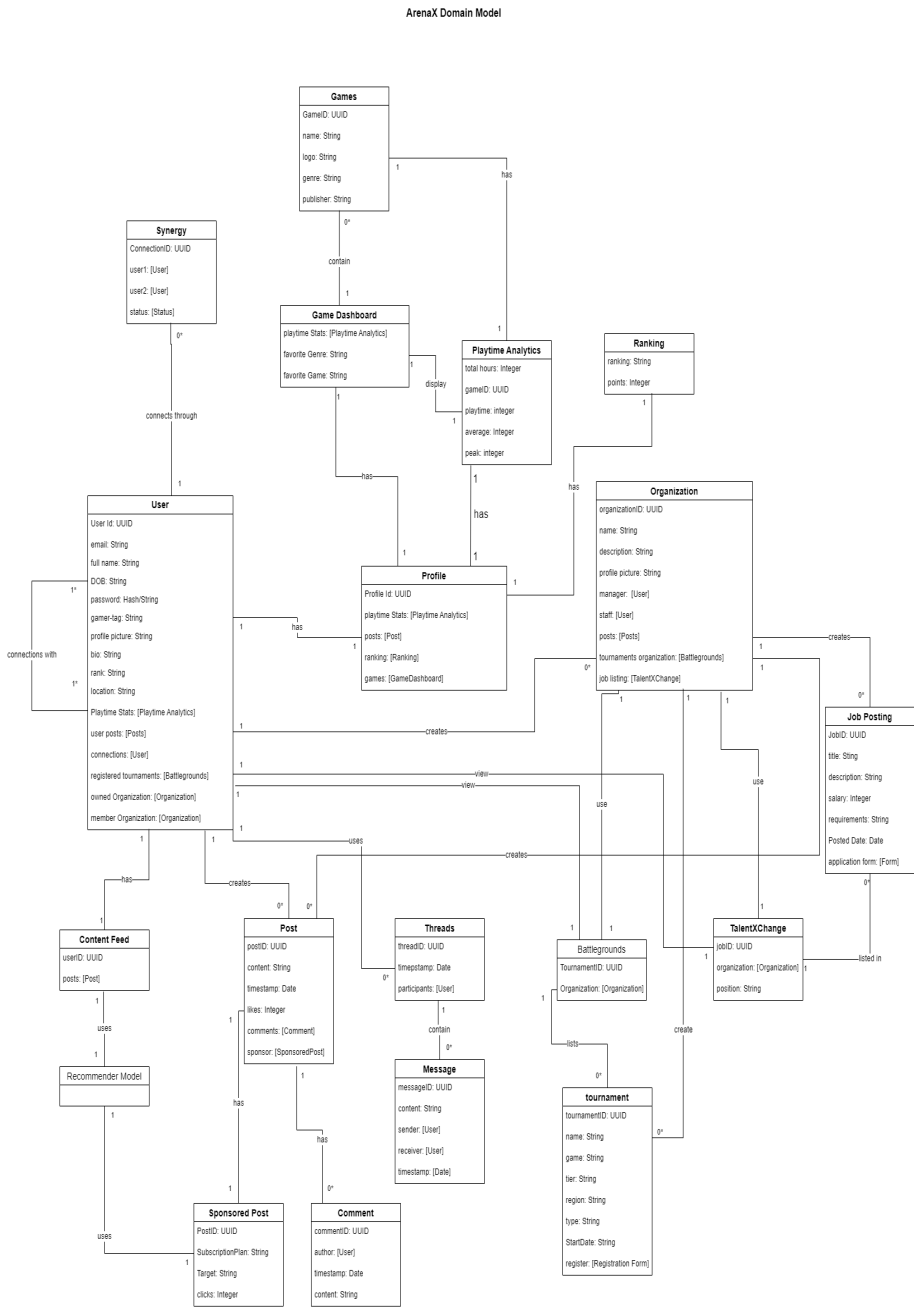


Figure 2.2: ArenaX Domain Model

Chapter 3

System Overview

ArenaX is a cross-platform networking app that provides a socio-professional environment for the gaming industry. It comprises of a personalized user feed system, interest based connections, threads, game-time tracking and analytics and tools for esports management. The product aims to converge the gaming community on a single platform by providing all necessary features and functionalities that attract gamers across genres.

3.1 Architectural Design

The ArenaX project is modularized based on distinct features to create a multi-layered architecture to ensure maintainability and scalability. The system is divided into five primary modules, each comprising of various features as sub-modules. These sub-modules are interconnected within their respective modules, creating a cohesive system. Each main feature within a module functions as an independent entity, ensuring the three-layered architecture of ArenaX, which consists of primary modules, sub-modules, and individual features within each sub-module. Below is an overview of the relationships between the primary modules of the system.

3.1.1 Modular Relationships

- **UserProfile:** Userprofile includes both module 1 and 2 of the app. Both offer same functionality but on different platforms. They collaborate with each other through the back-end server setup. These modules focuses mainly on user profile management, game-time tracking and game-dashboard management. These modules are related to all the other modules that leverage the recommender system in their functionalities. These modules provide the userdata to predict user interest and recommend content and connections, a feature of the Networking module. Similarly, the

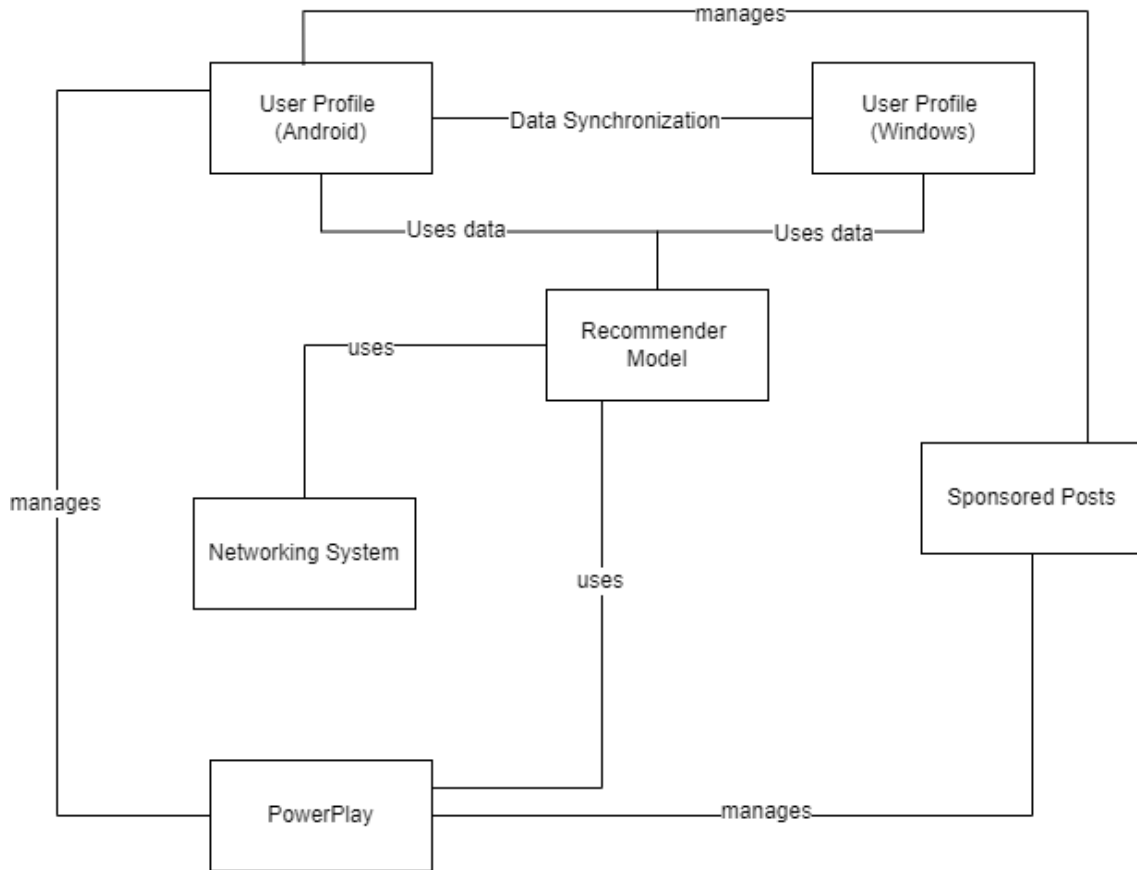


Figure 3.1: Modular Relationships

user profile modules provide user interests for the Powerplay module to maintain interest based Battlegrounds and TalentXchange lists. In conclusion, these modules mainly collaborate with the rest of modules to feed them userdata to cater users' interest-based features of those modules.

- **Networking System:** This module focuses on the user-networking part of the app and consists of features including personalized user feed, Synergy and Chats. This module relies on the data of the first module while it has no major collaboration with the third and fourth module of the app.
- **PowerPlay :** This is the most extensive module of the app consisting of all the features related to esports management tools including organization management, recruitment, staff management and event organization. This module collaborates with the user profile modules by making the Battlegrounds and TalentXchange sections of the app accessible for users.
- **PowerPlay :** This module is mainly independent and focuses on post sponsorships and advertisements. This module collaborates with the networking system to in-

crease the reach of a post within a target audience that has interest relevant to the post.

3.1.2 Architectural Pattern

The "Layered" architectural style is followed in the development of the Arenax project. It ensures modularity and efficient maintainability in addition to separating the presentation, business logic and database layers. Following is a detailed architecture diagram of the project.

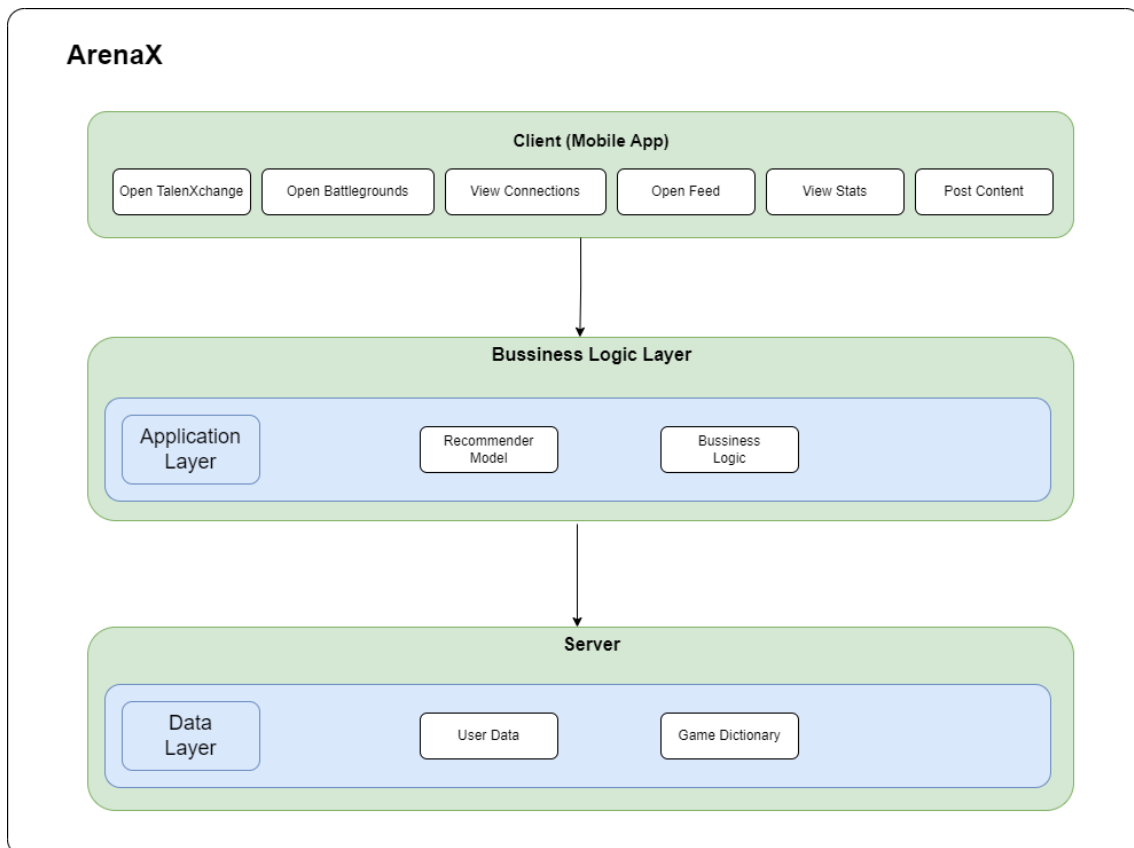


Figure 3.2: Architecture Diagram

3.2 Design Models

Following are the design models of the system.

3.2.1 Activity Diagram

1. UC 01 - Register User:

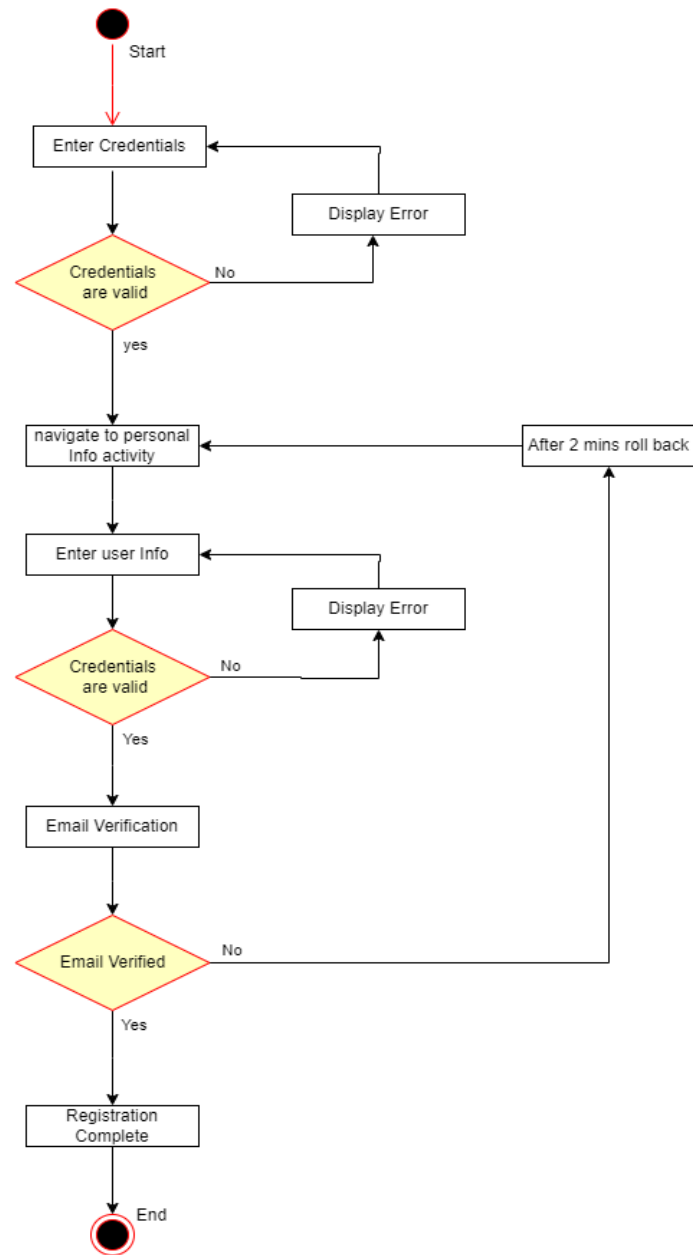


Figure 3.3: UC 01 - Register User

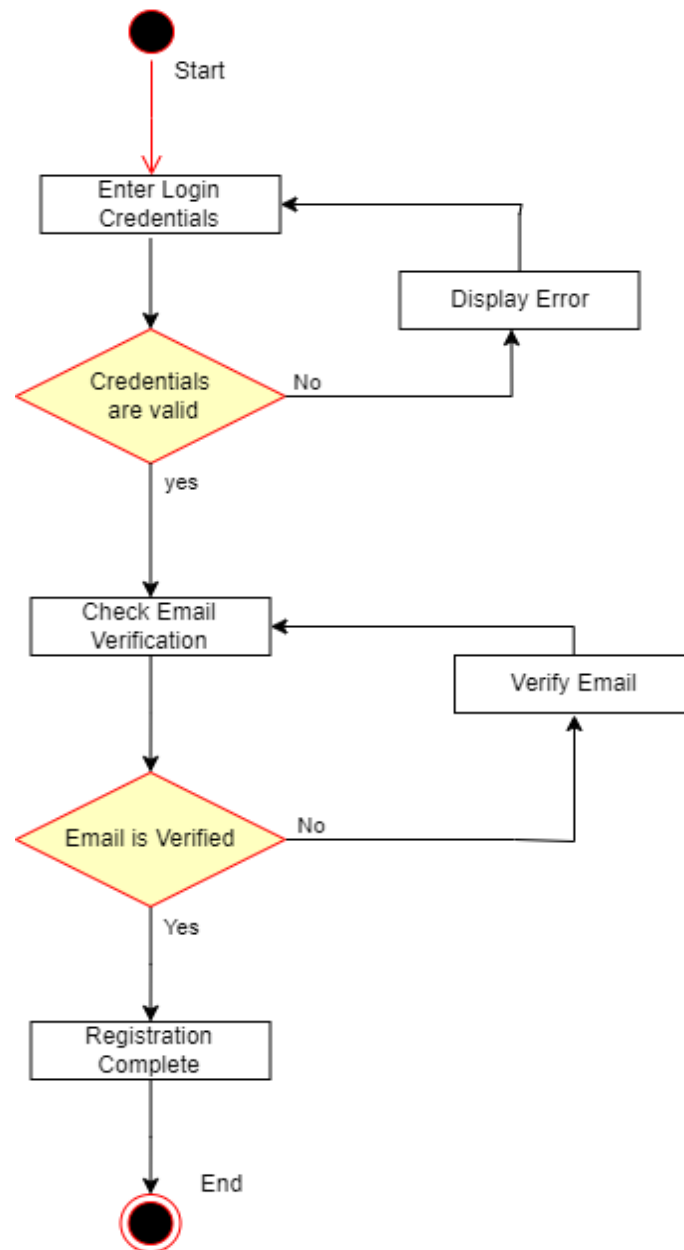
2. UC 02 - User Login:

Figure 3.4: UC 02 - User Login

3. UC 03 - Manage User Profile:

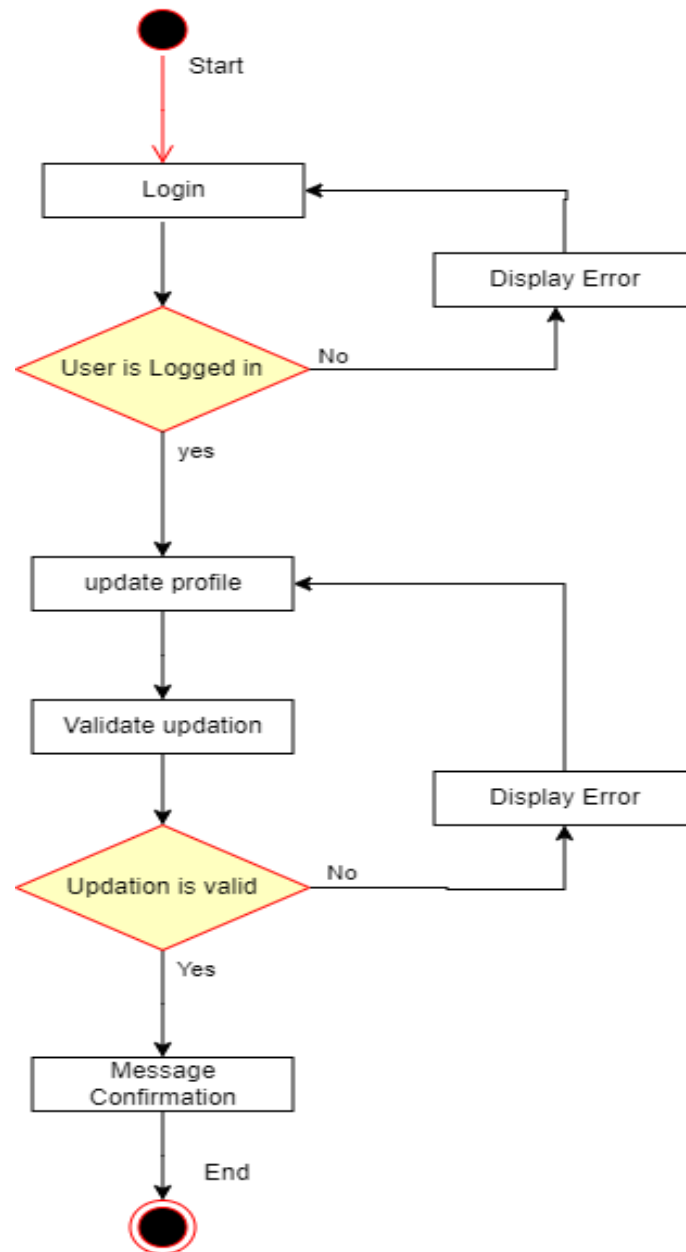


Figure 3.5: UC 03 - Manage User Profile

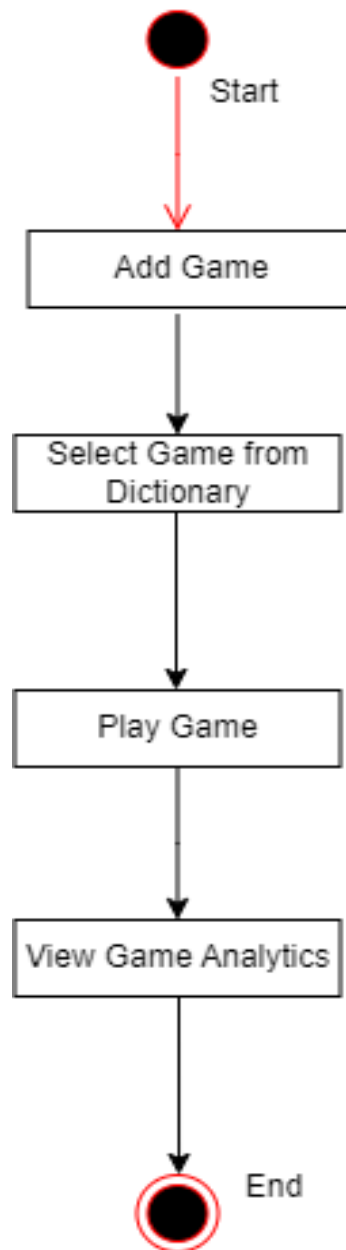
4. UC 04 - Manage Games Dashboard:

Figure 3.6: UC 04 - Manage Games Dashboard

5. UC 05 - View Game Analytics:

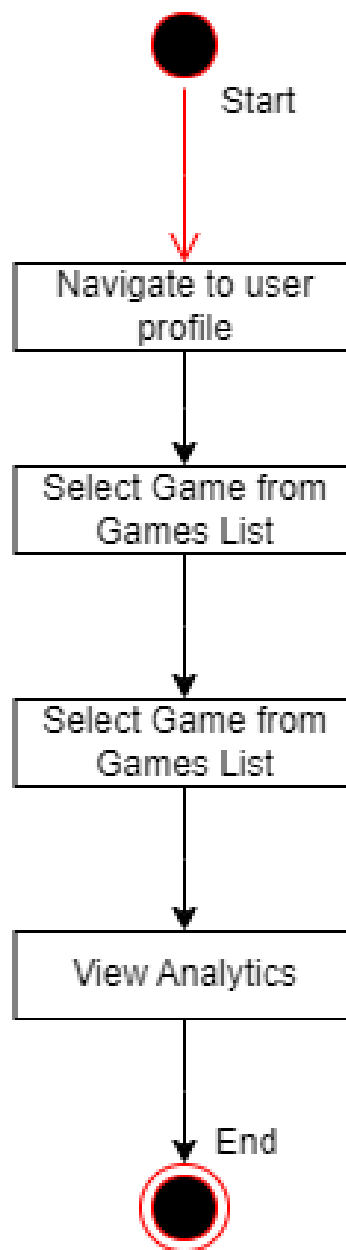


Figure 3.7: UC 05 - View Game Analytic

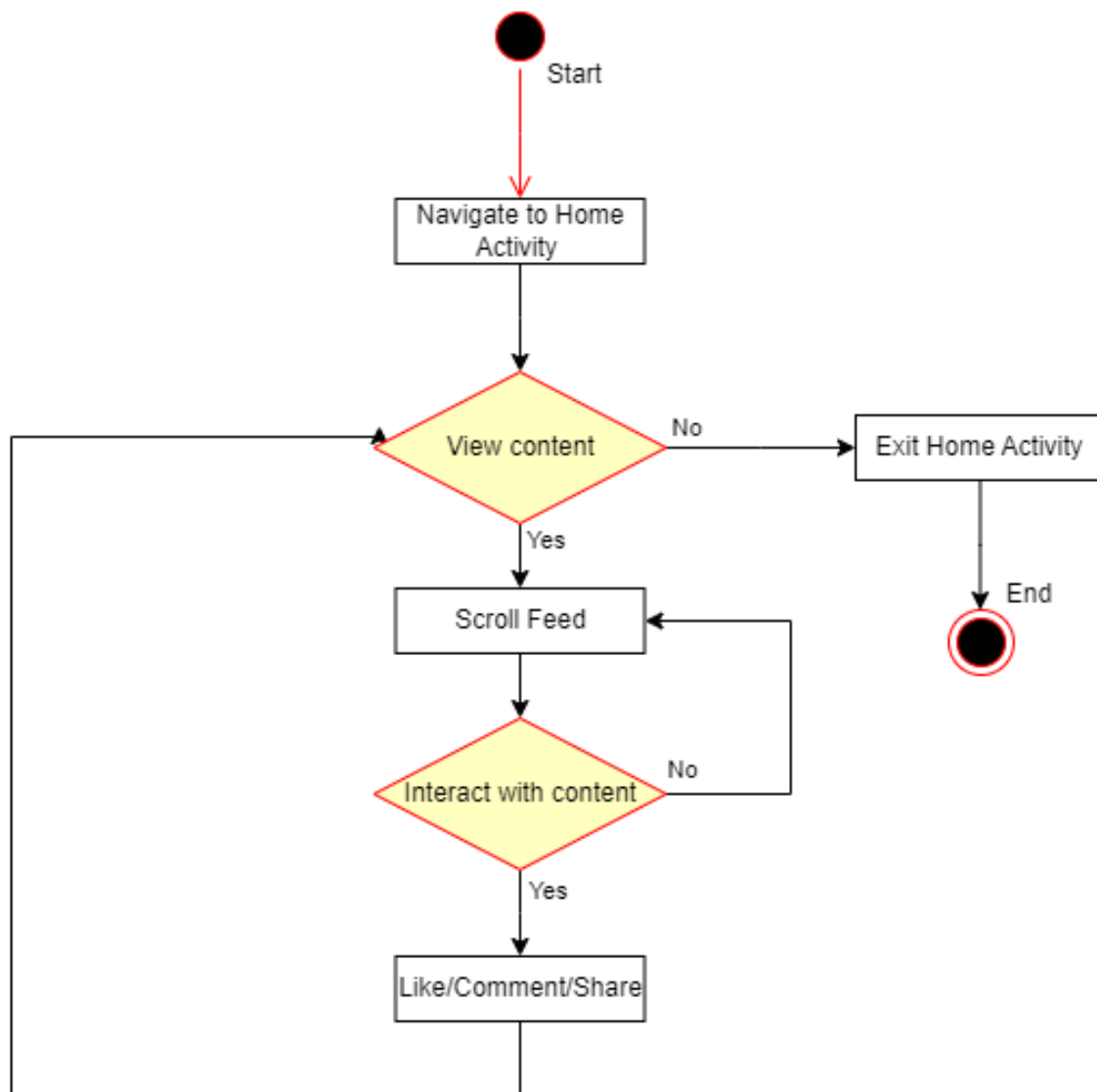
6. UC 06 - View User Feed:

Figure 3.8: UC 06 - View User Feed

7. UC 07 - Manage Connections:

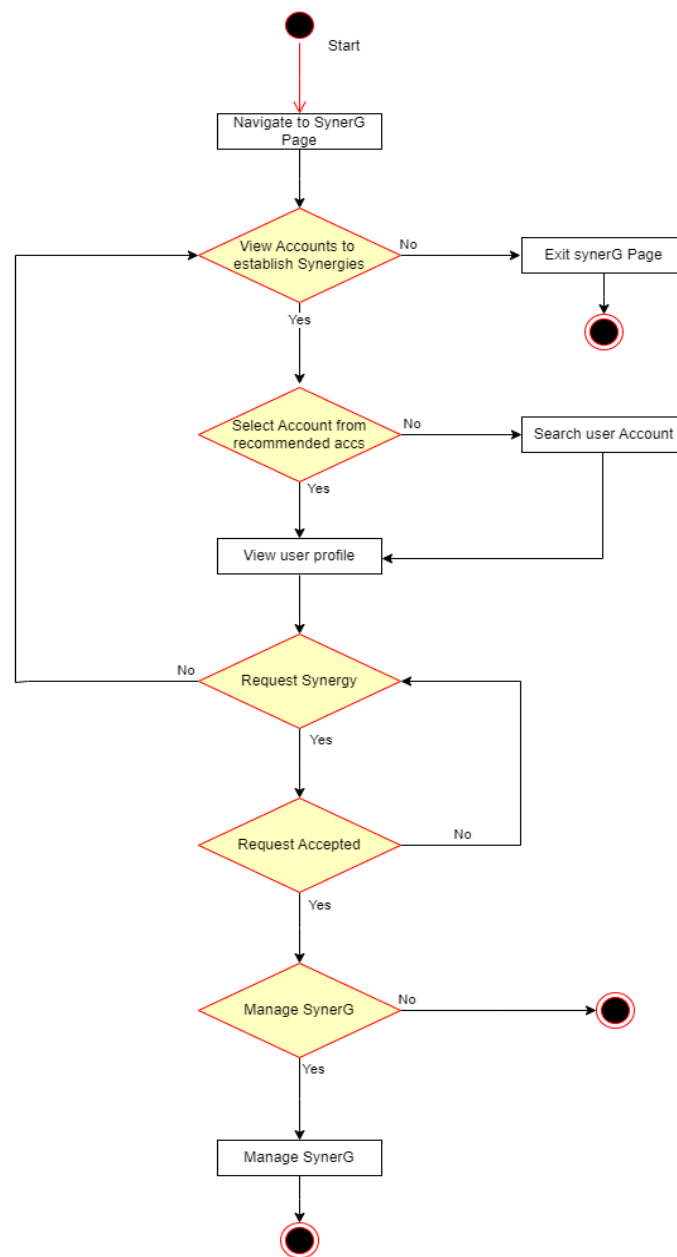


Figure 3.9: UC 07 - Manage Connections

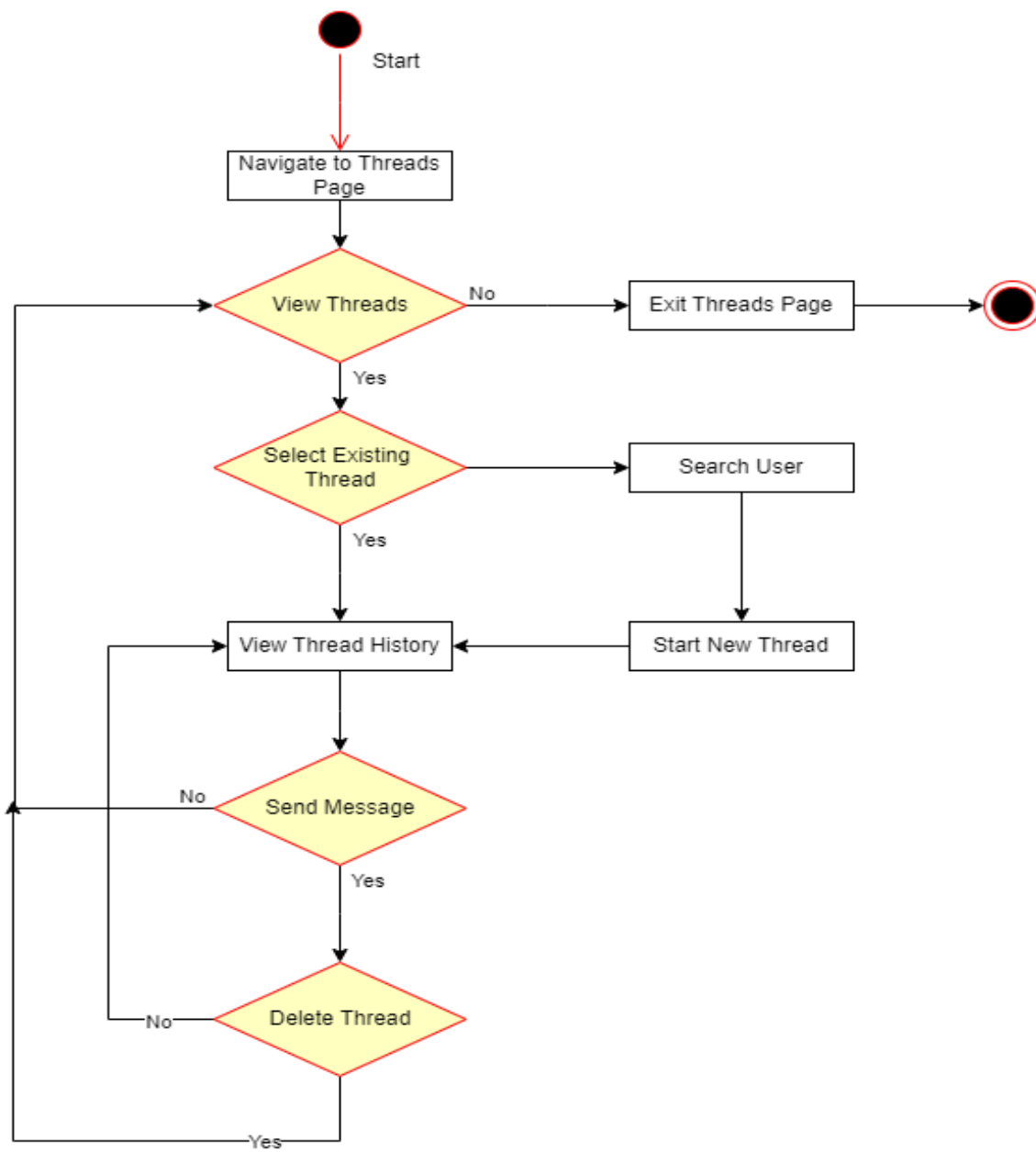
8. UC 08 - Manage Threads:

Figure 3.10: UC 08 - Manage Threads

9. UC 09 - Manage Organization Profile:

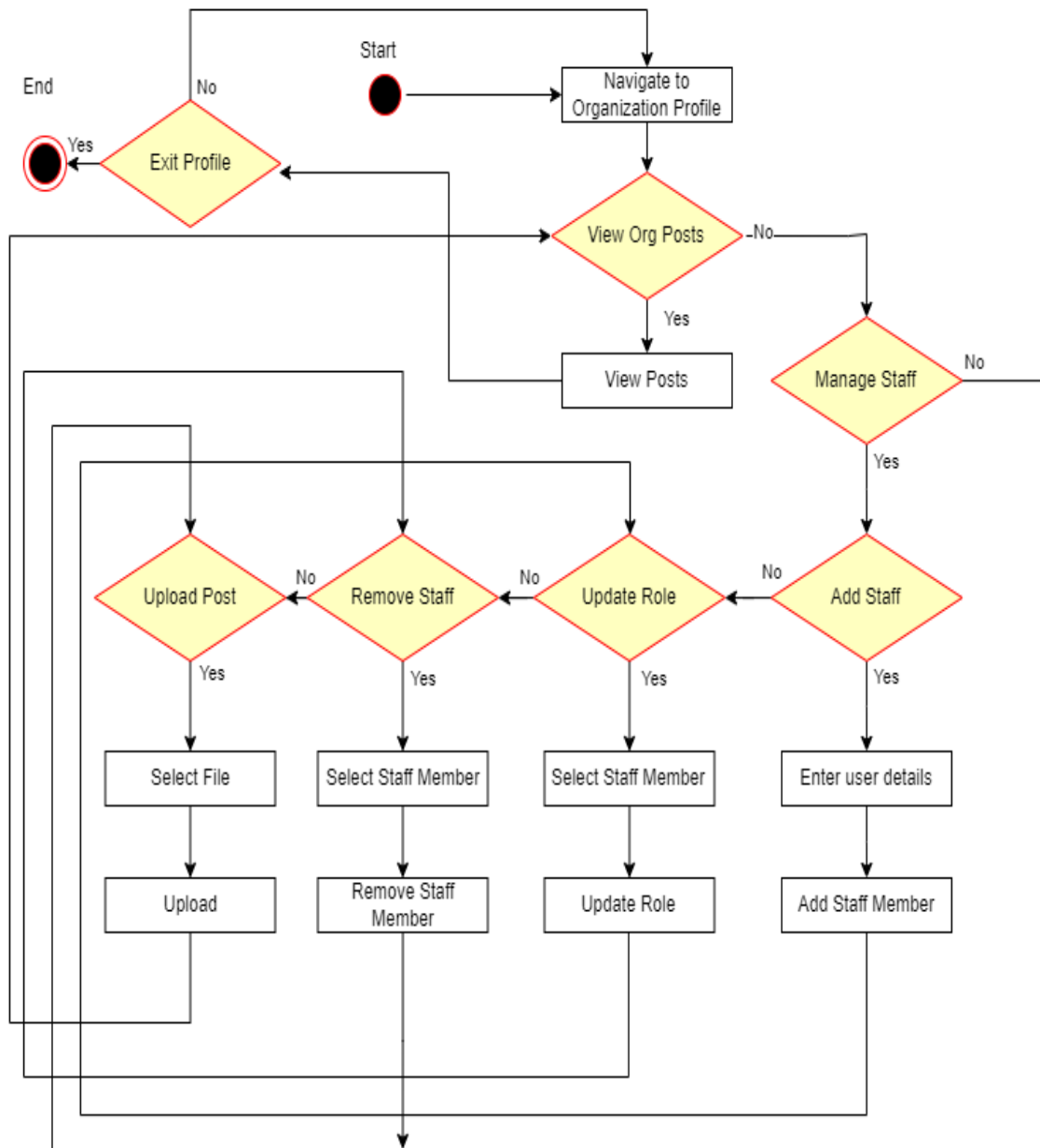


Figure 3.11: UC 09 - Manage Organization Profile

10. UC 10 - Manage Battlegrounds:

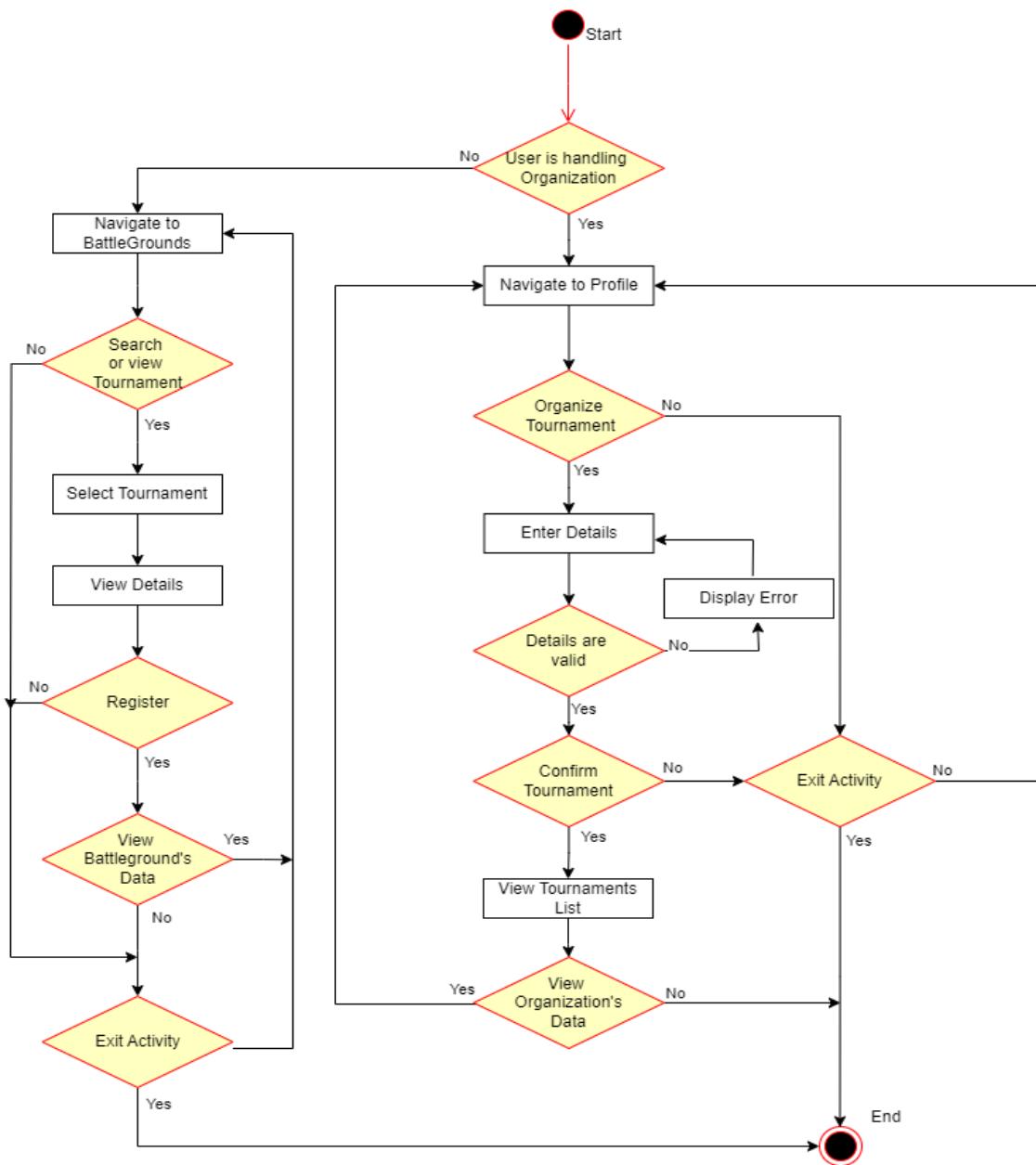


Figure 3.12: UC 10 - Manage Battlegrounds

11. UC 11 - Manage TalentXchange:

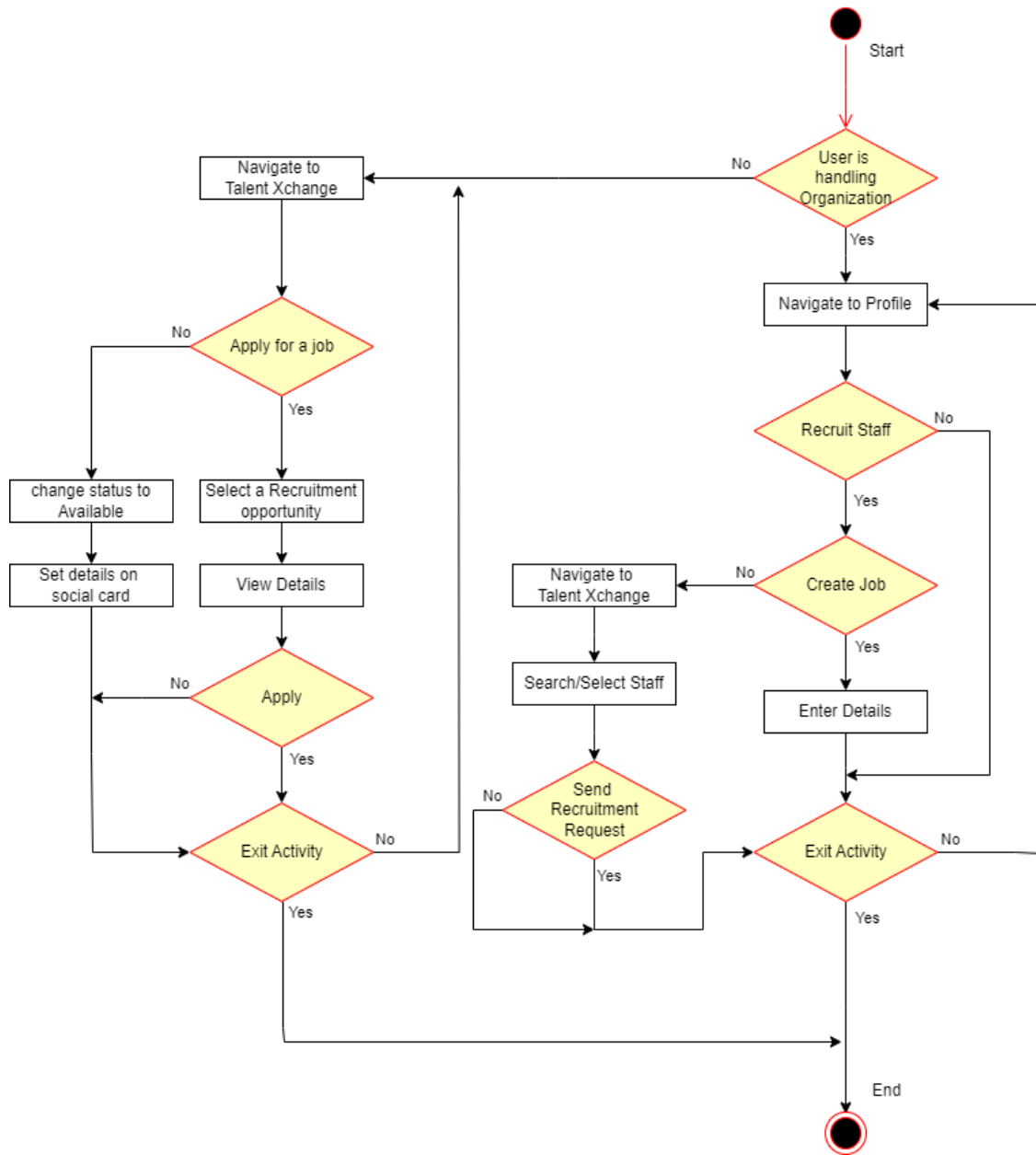


Figure 3.13: UC 11 - Manage TalentXchange

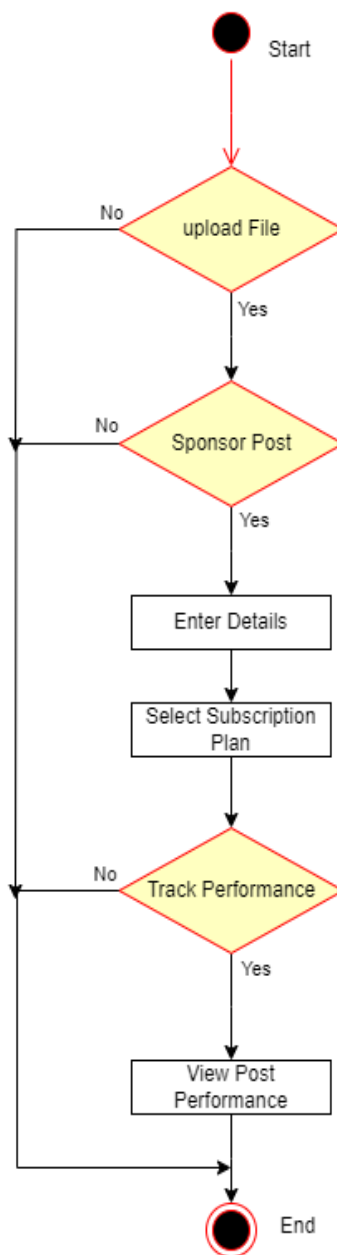
12. UC 12 - Manage Sponsored Posts:

Figure 3.14: UC 12 - Manage Sponsored Posts

3.2.2 Class Diagram

Below depicted is an extensive class diagram of the app.

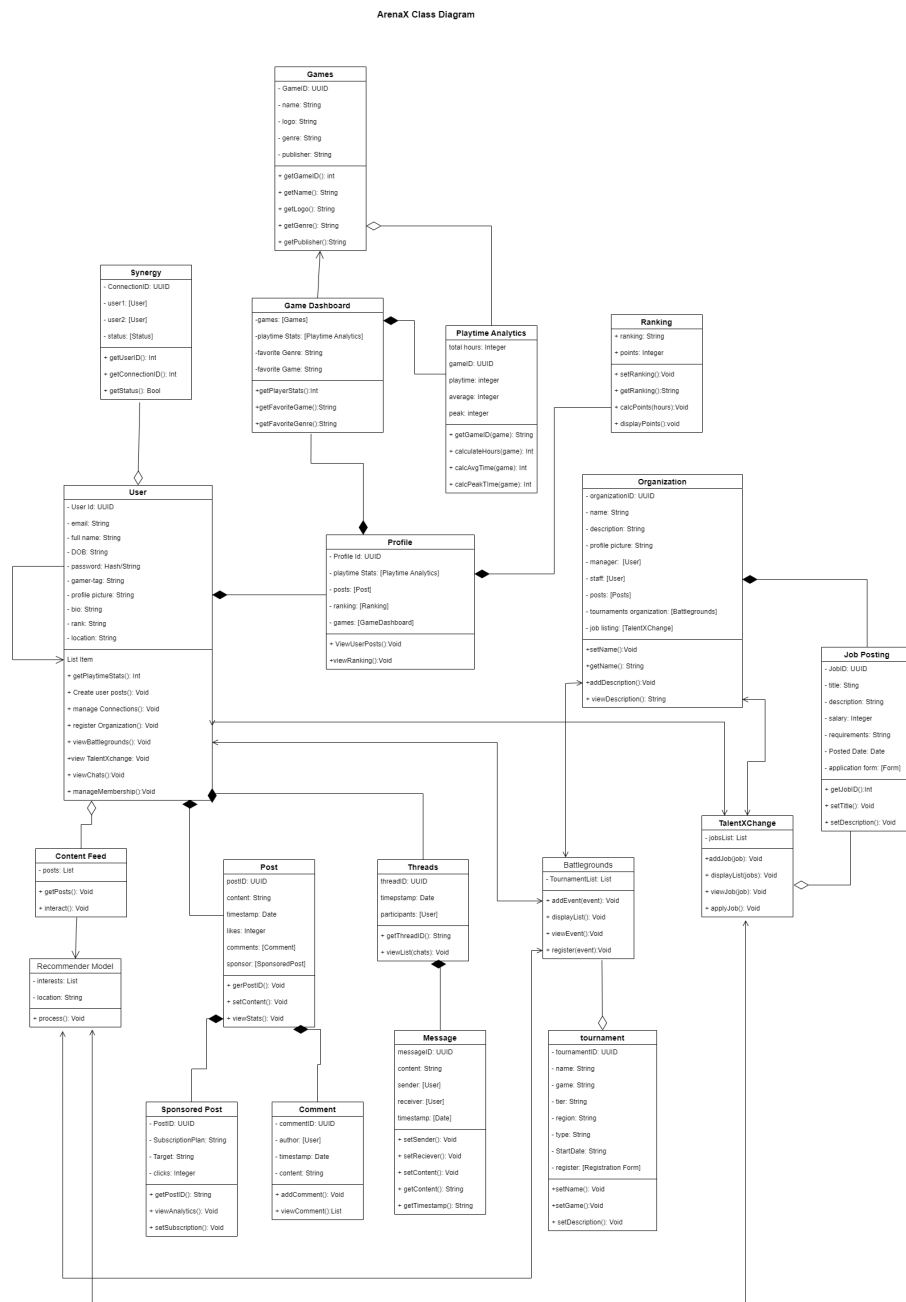


Figure 3.15: ArenaX Class Diagram

3.2.3 Class-level Sequence Diagram

Following are the class-level sequence diagram of each use case.

1. UC 01 - Register User:

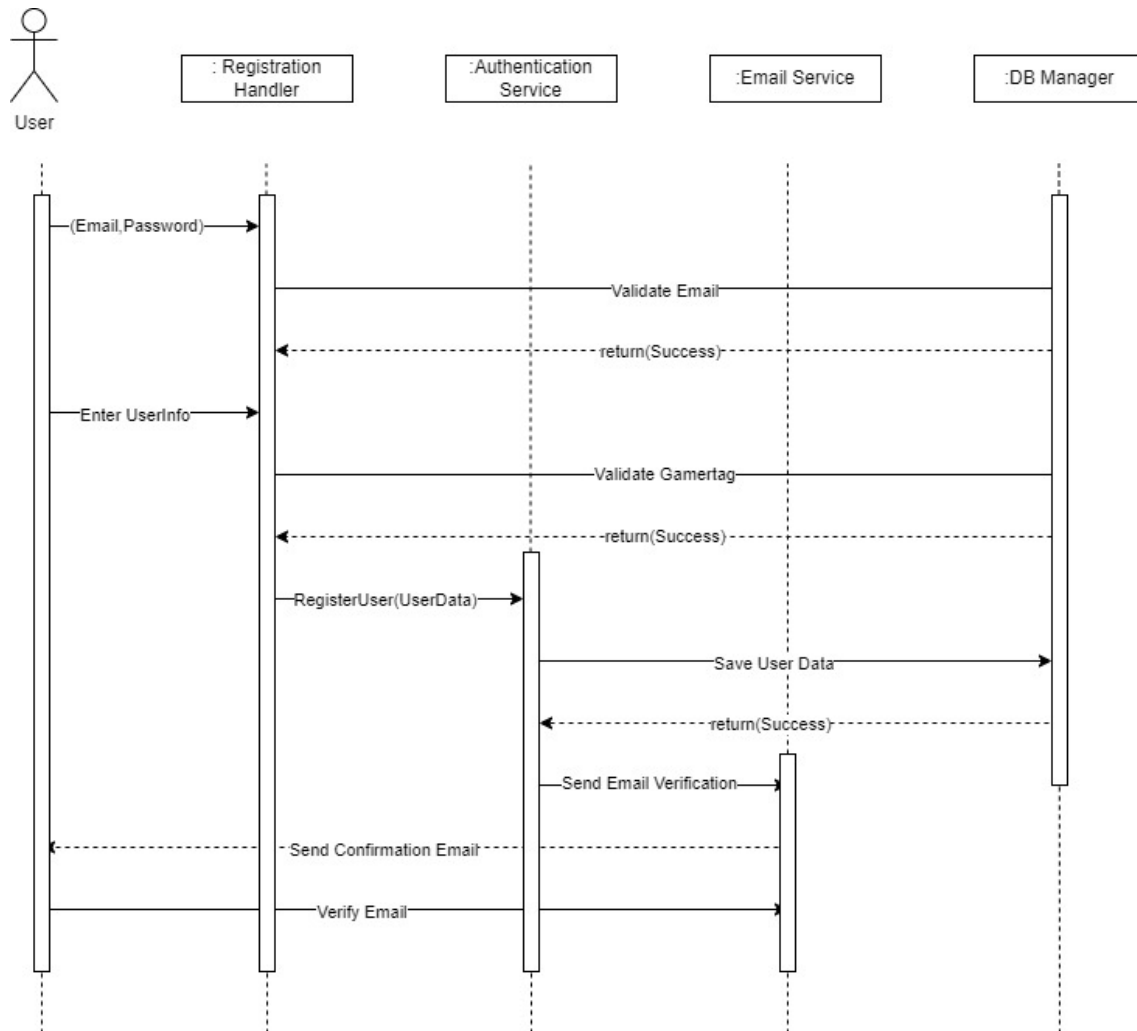


Figure 3.16: UC 01 - Register User

2. UC 02 - User Login:

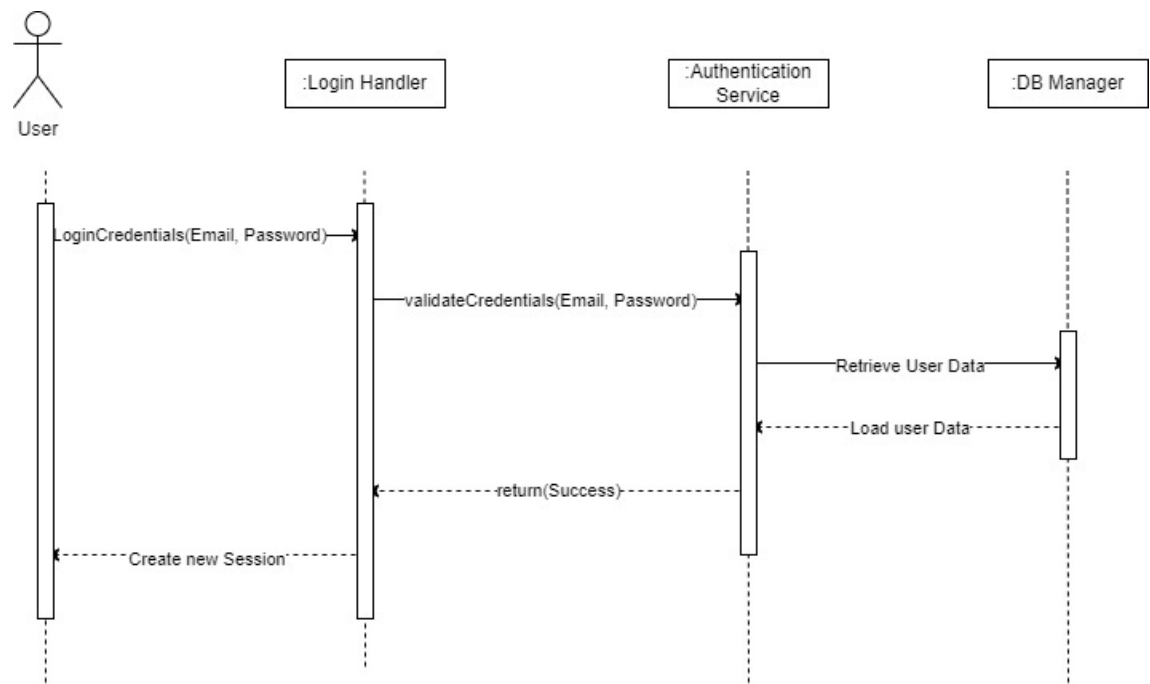


Figure 3.17: UC 02 - User Login

3. UC 03 - Manage User Profile:

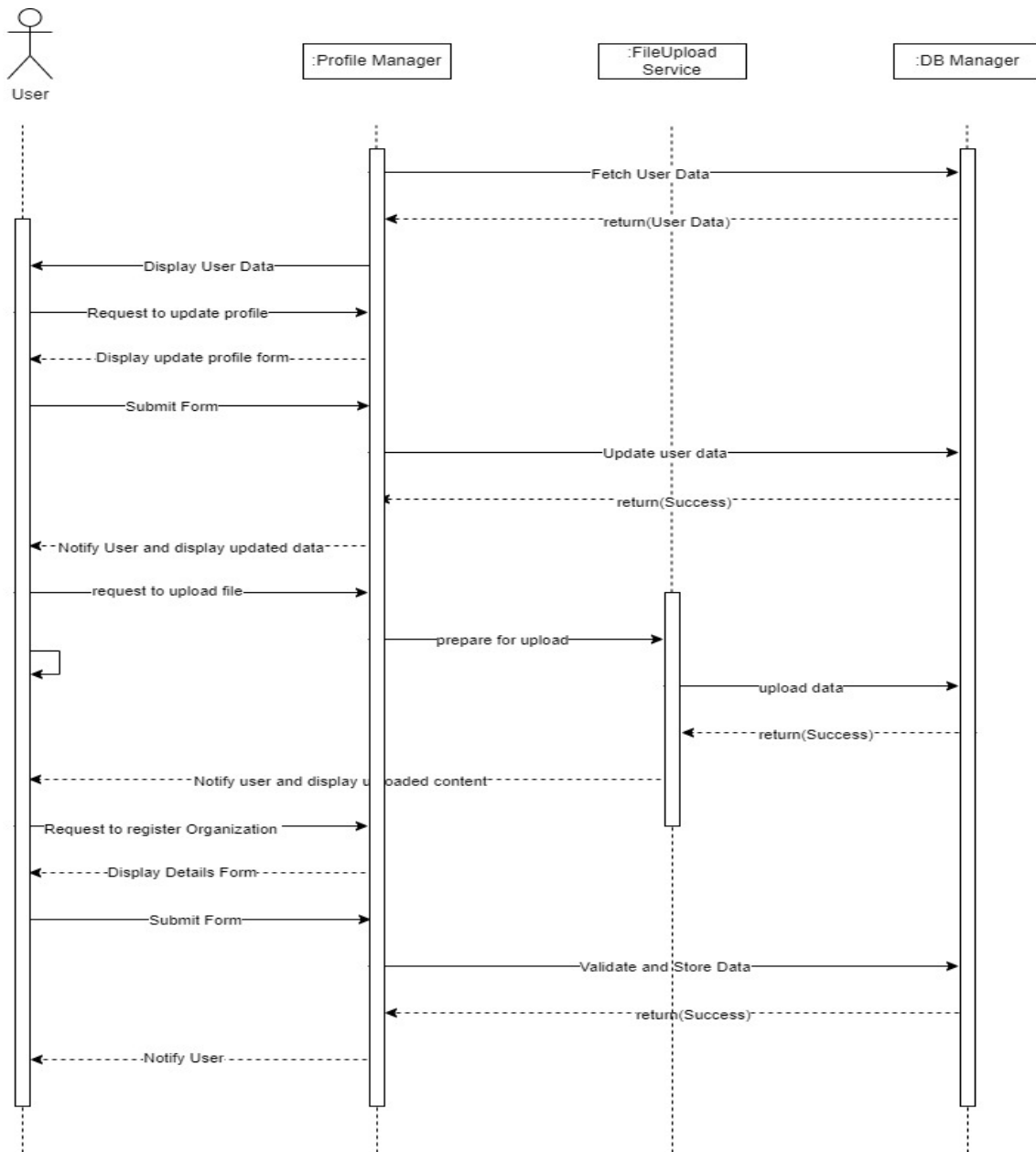


Figure 3.18: UC 03 - Manage User Profile

4. UC 04 - Manage Games Dashboard:

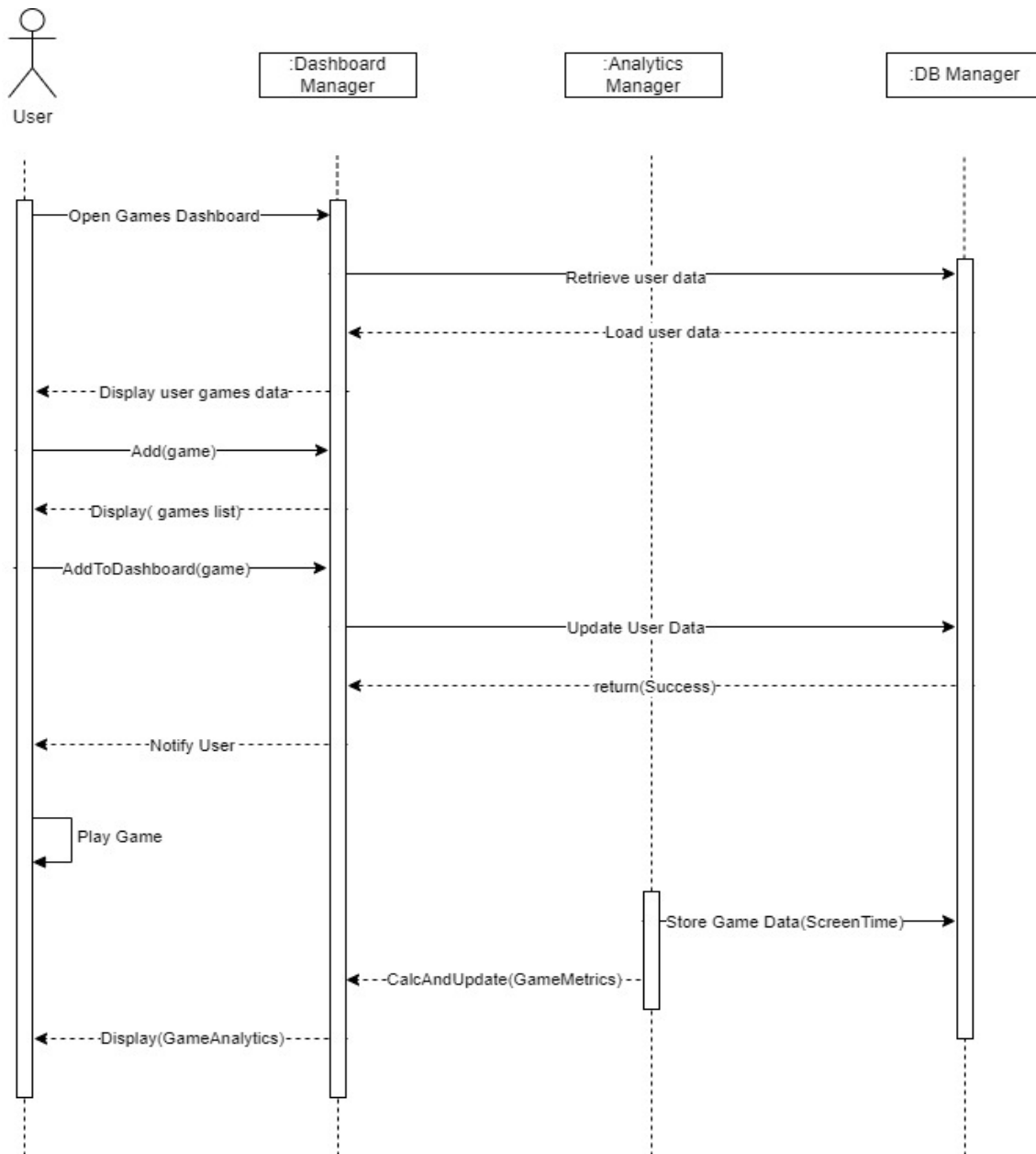


Figure 3.19: UC 04 - Manage Games Dashboard

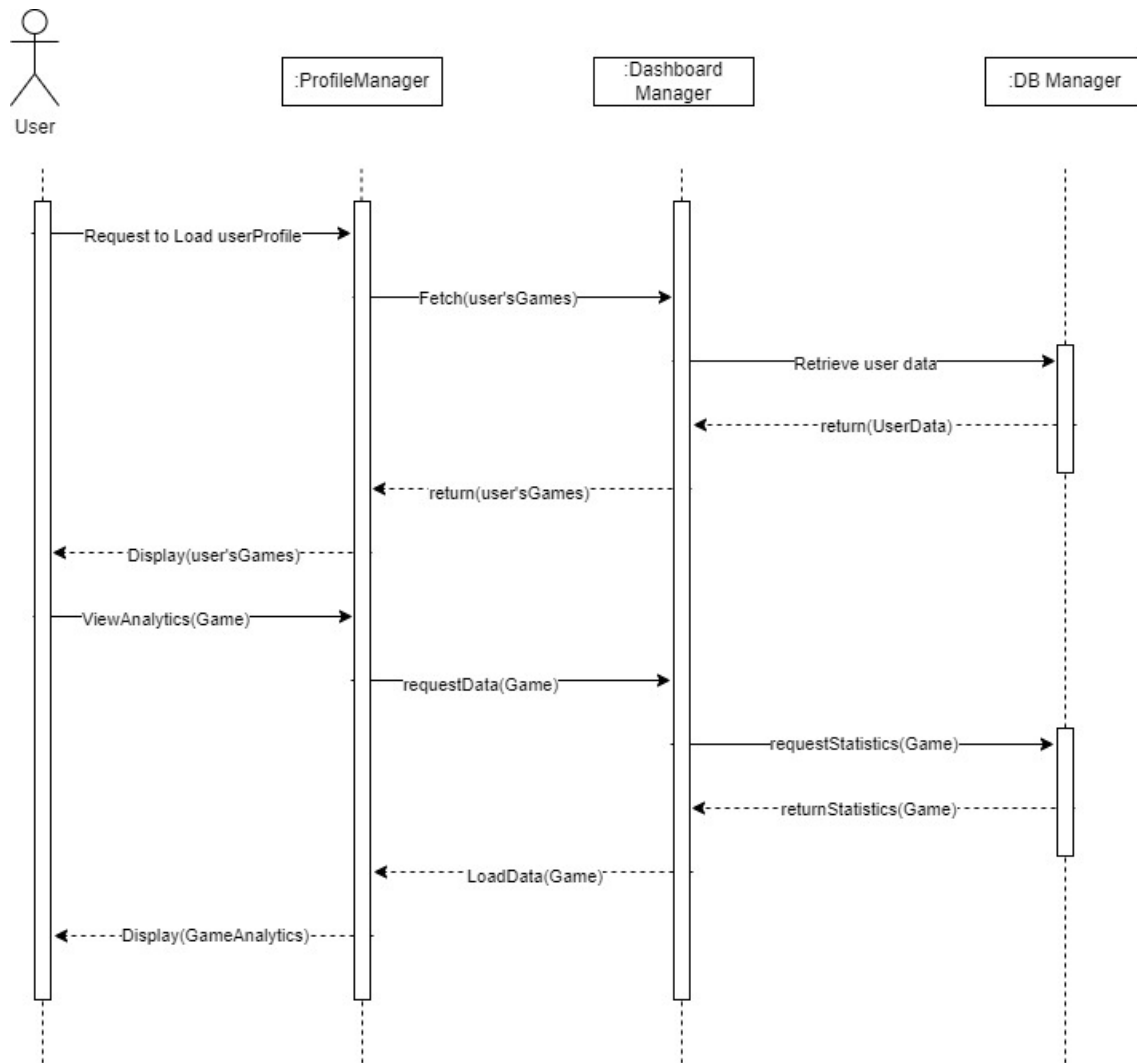
5. UC 05 - View Game Analytics:

Figure 3.20: UC 05 - View Game Analytic

6. UC 06 - View User Feed:

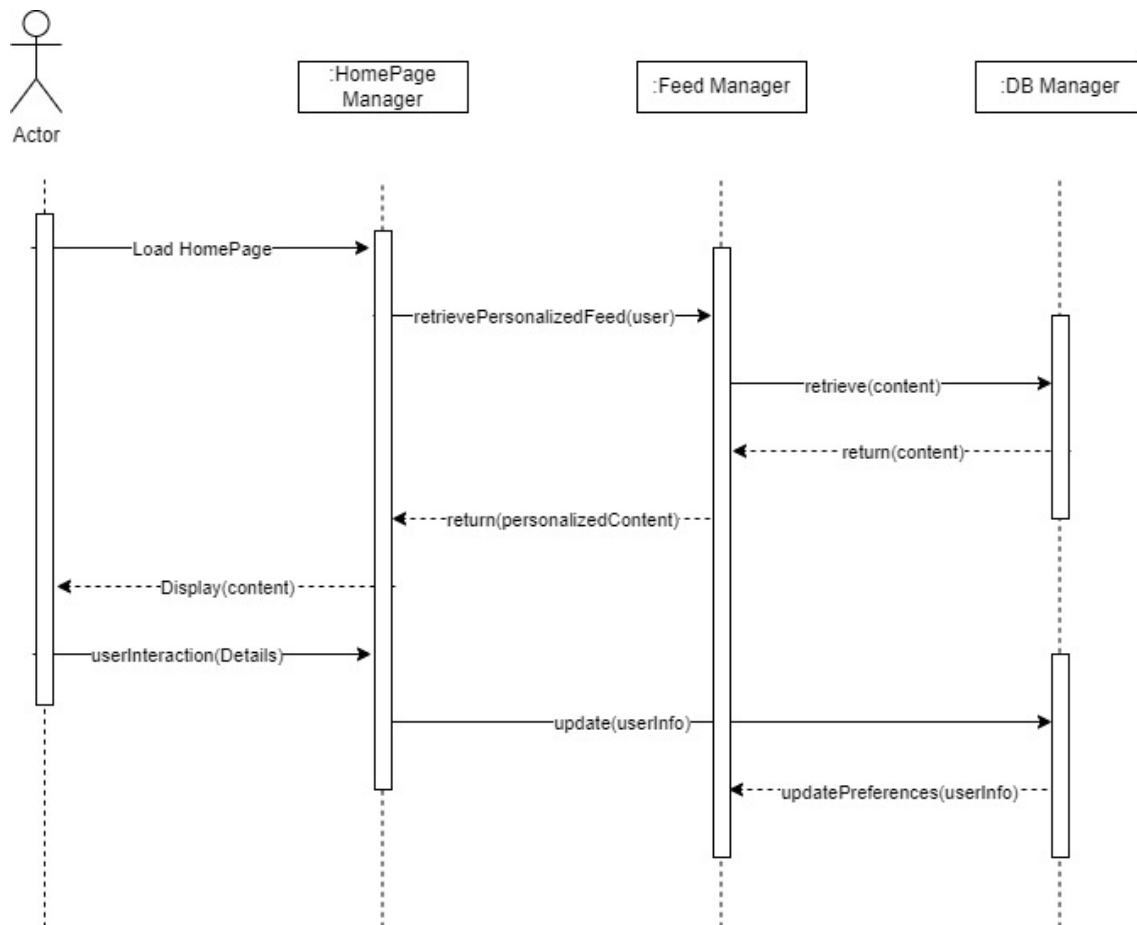


Figure 3.21: UC 06 - View User Feed

7. UC 07 - Manage Connections:

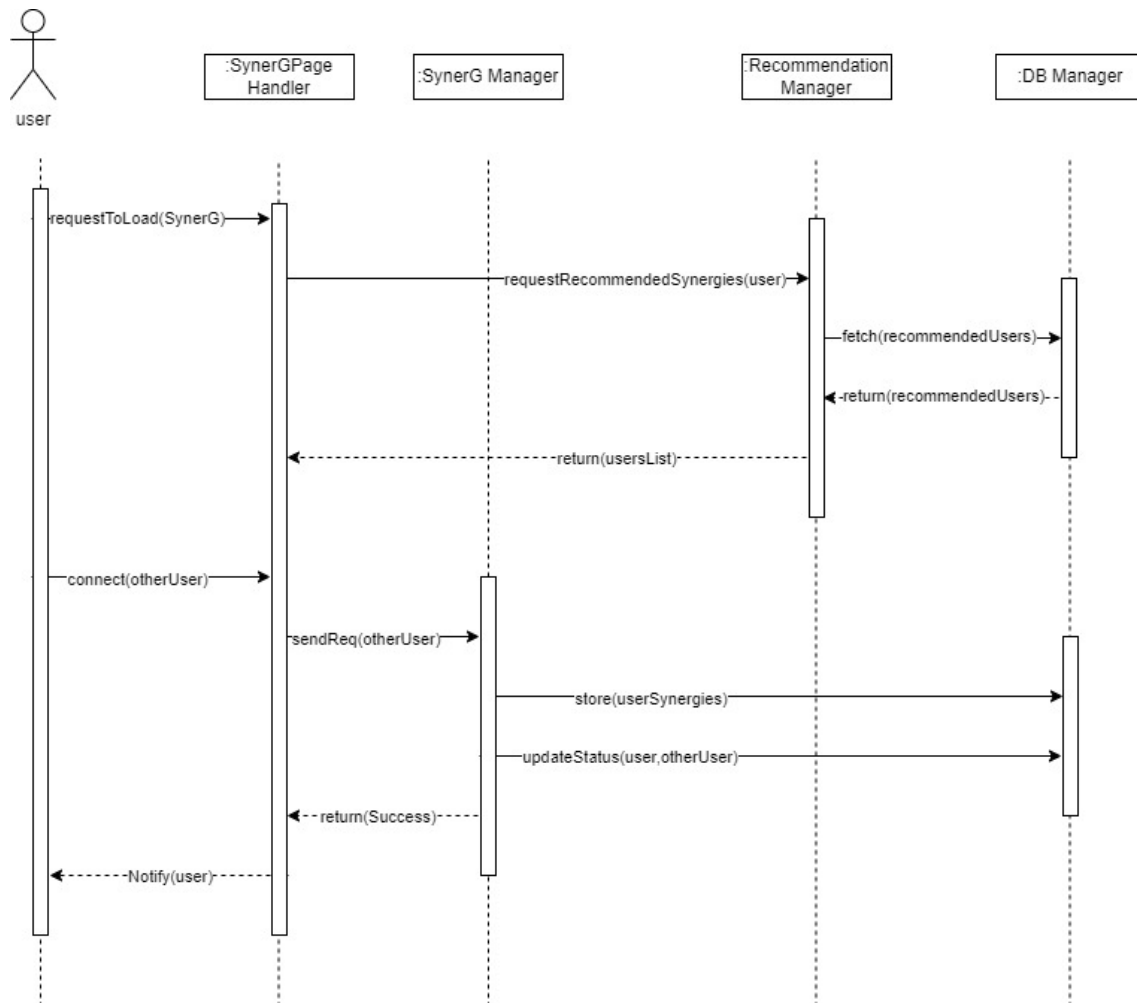


Figure 3.22: UC 07 - Manage Connections

8. UC 08 - Manage Threads:

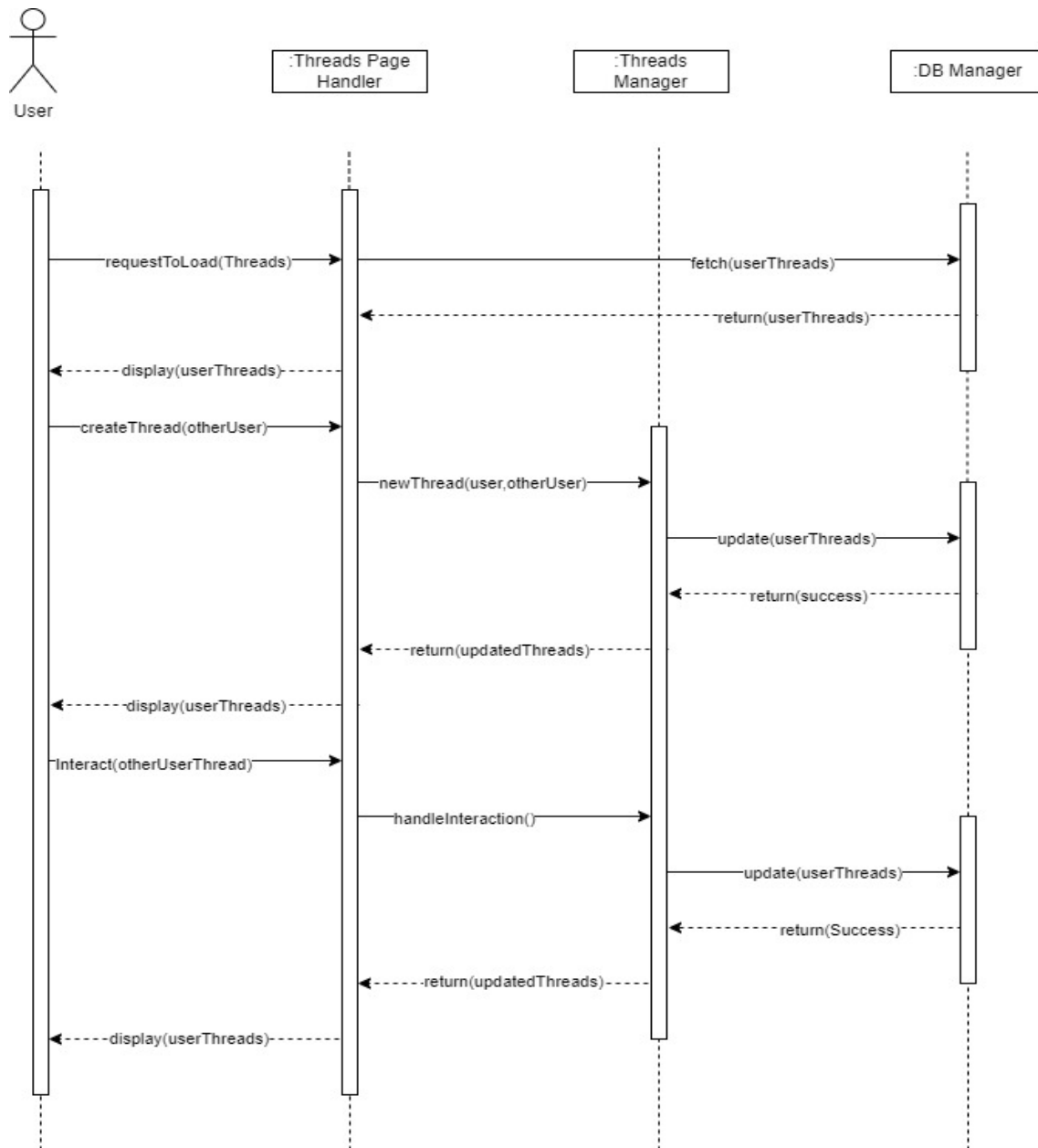


Figure 3.23: UC 08 - Manage Threads

9. UC 09 - Manage Organization Profile:

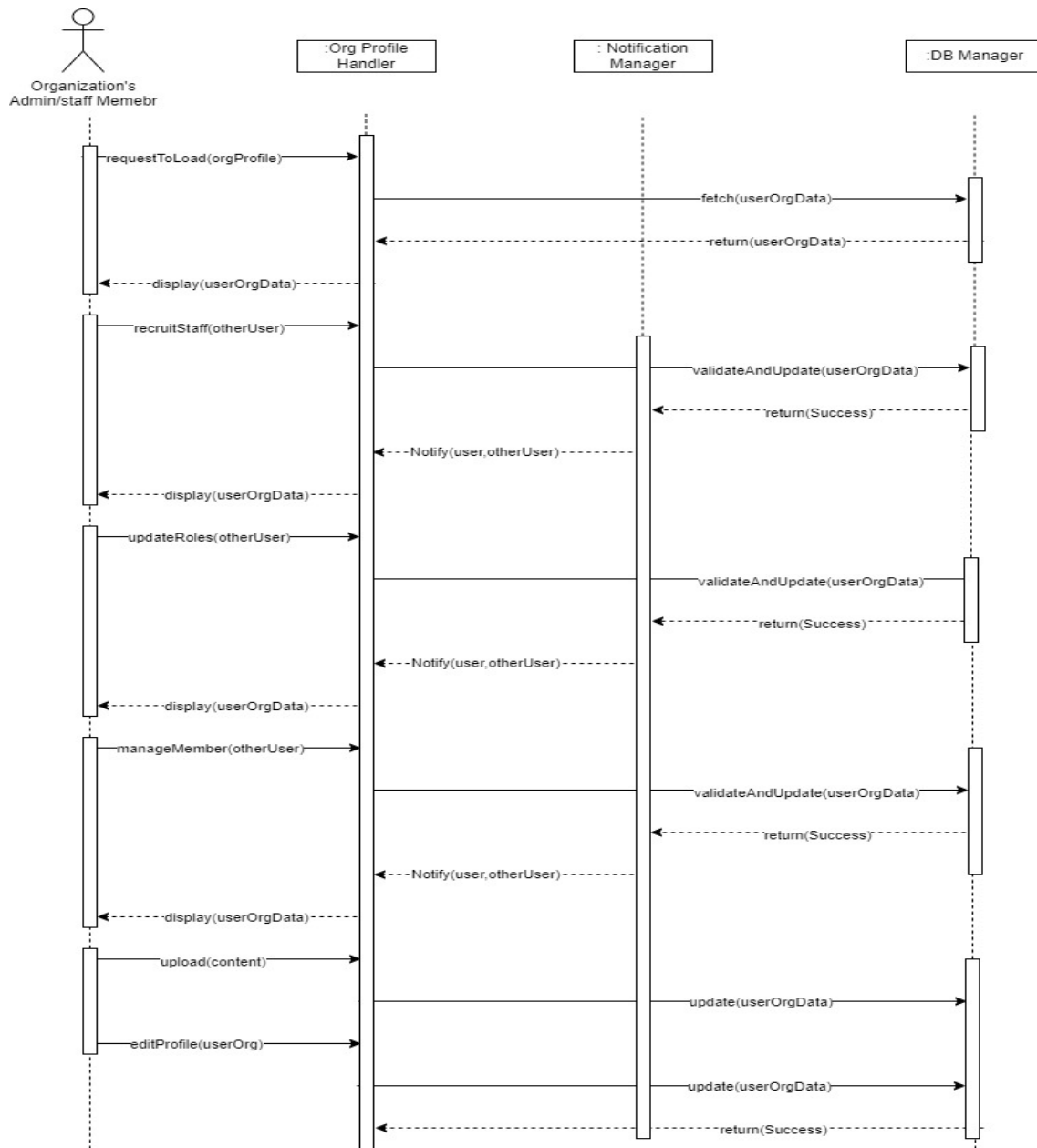


Figure 3.24: UC 09 - Manage Organization Profile

10. UC 10 - Manage Battlegrounds:

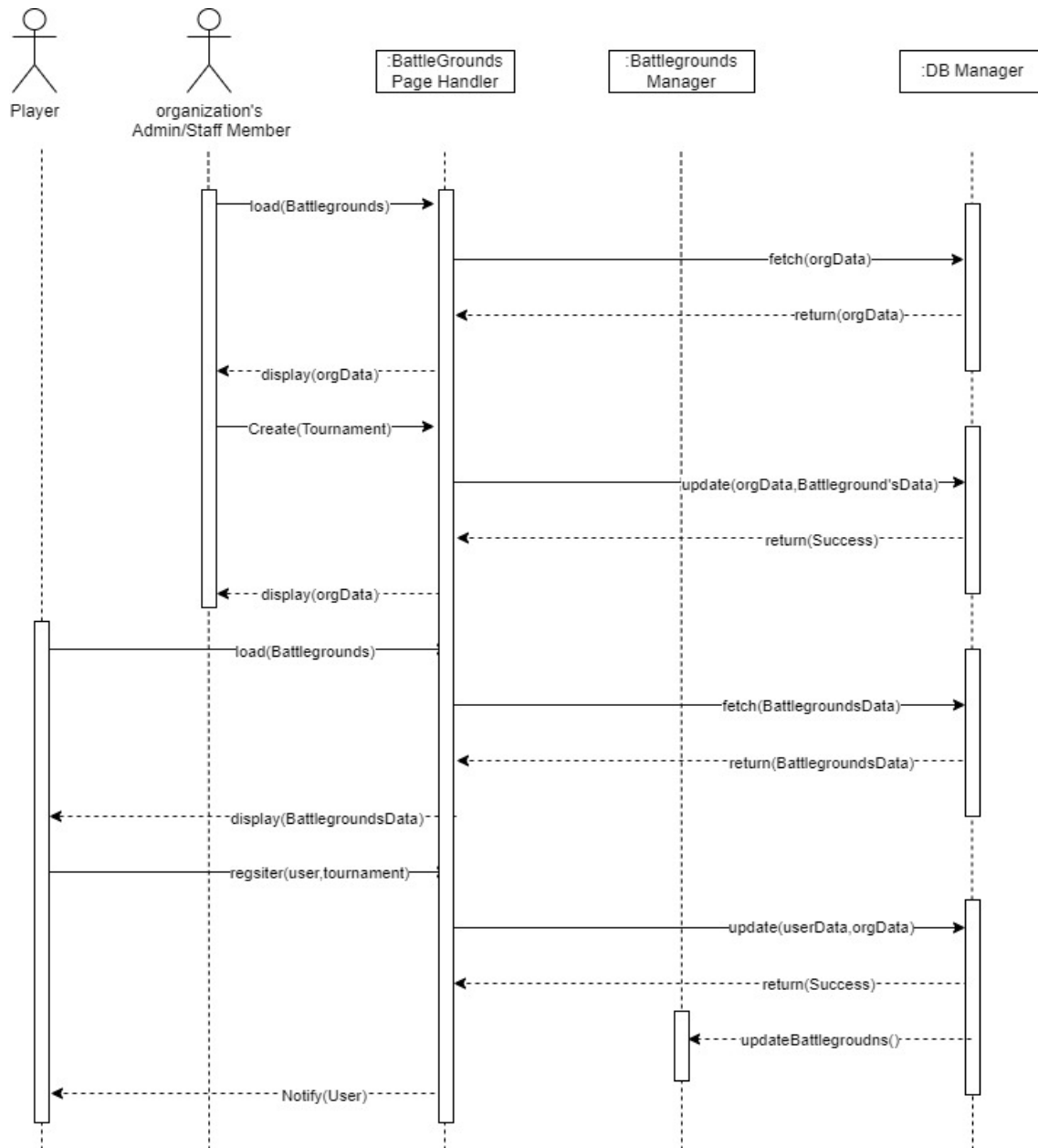


Figure 3.25: UC 10 - Manage Battlegrounds

11. UC 11 - Manage TalentXchange:

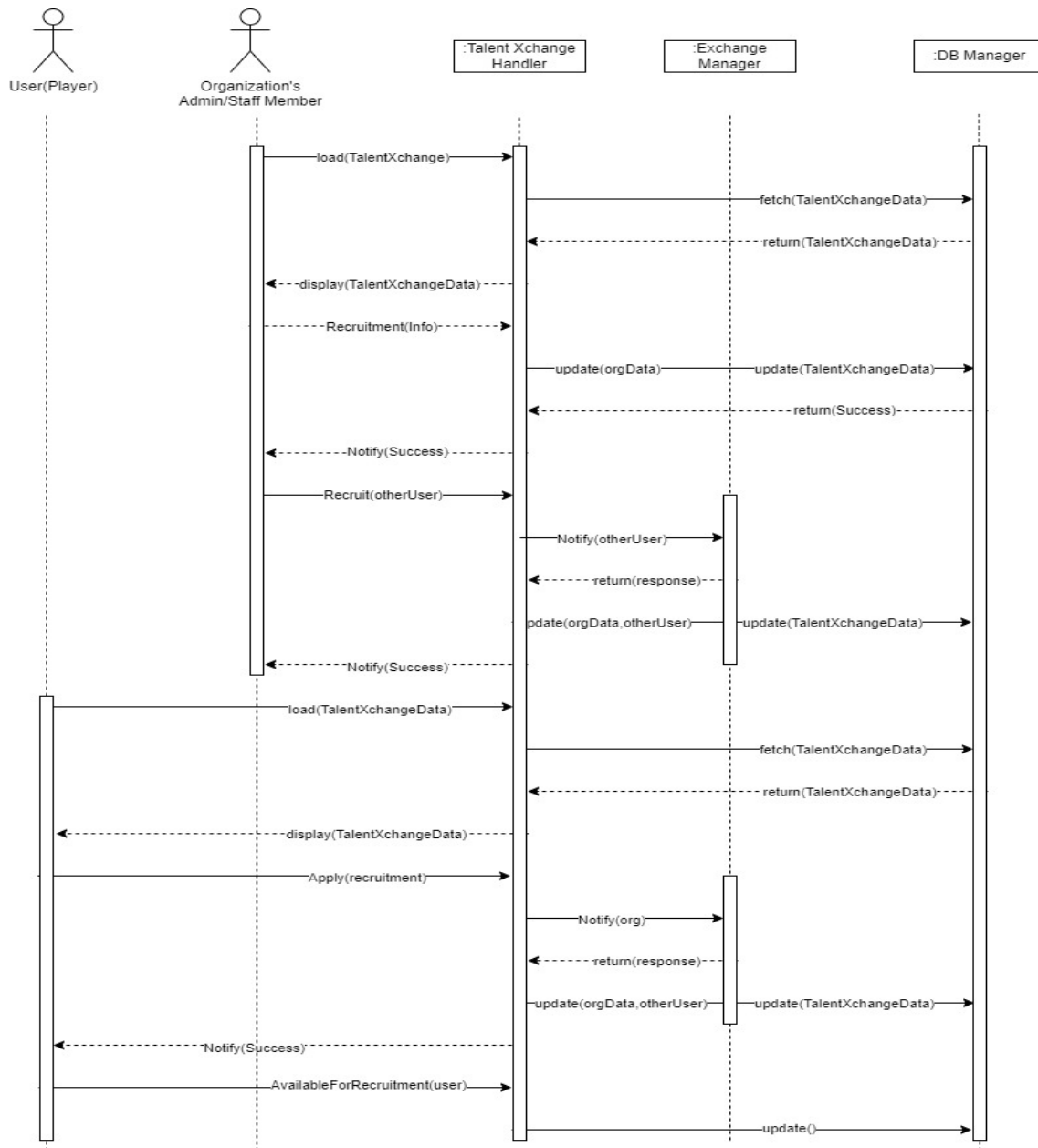


Figure 3.26: UC 11 - Manage TalentXchange

12. UC 12 - Manage Sponsored Posts:

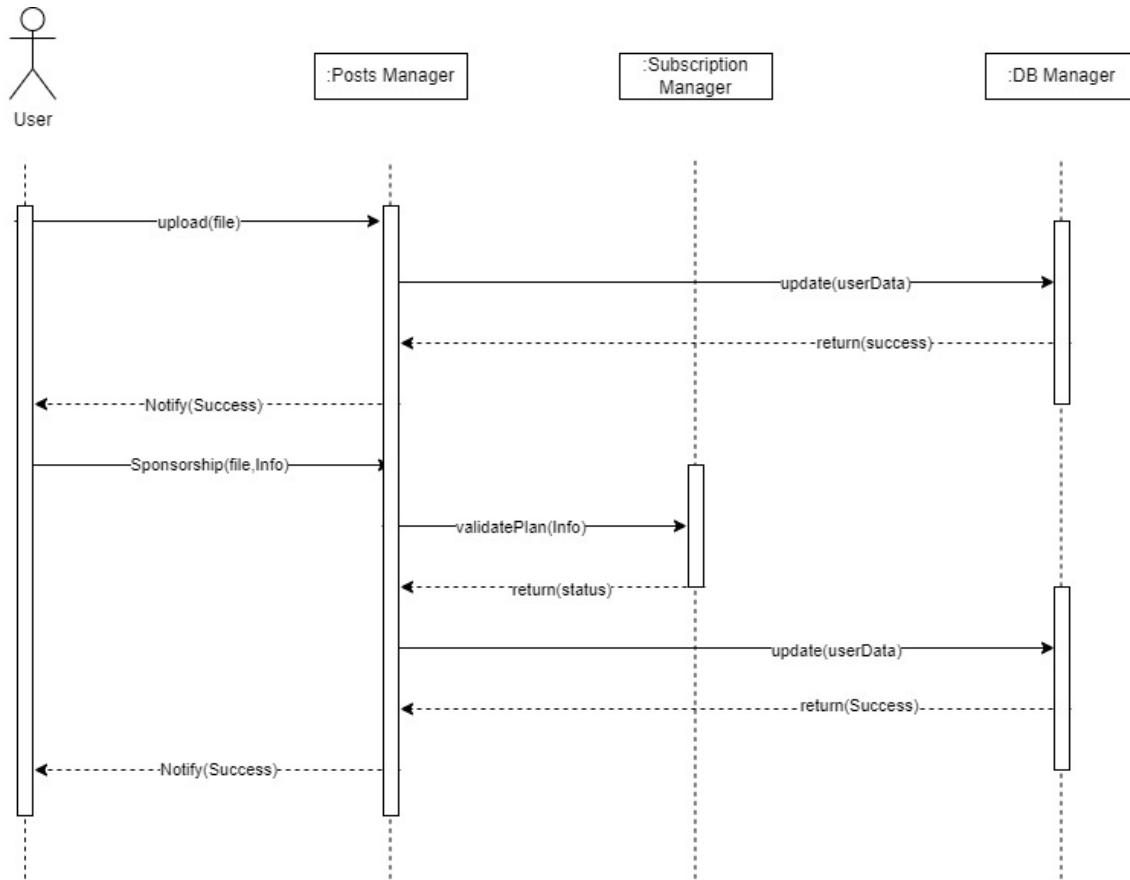


Figure 3.27: UC 12 - Manage Sponsored Posts

3.2.4 State Transition Diagrams

Following are the State transition diagrams of the product.

1. User Profile:

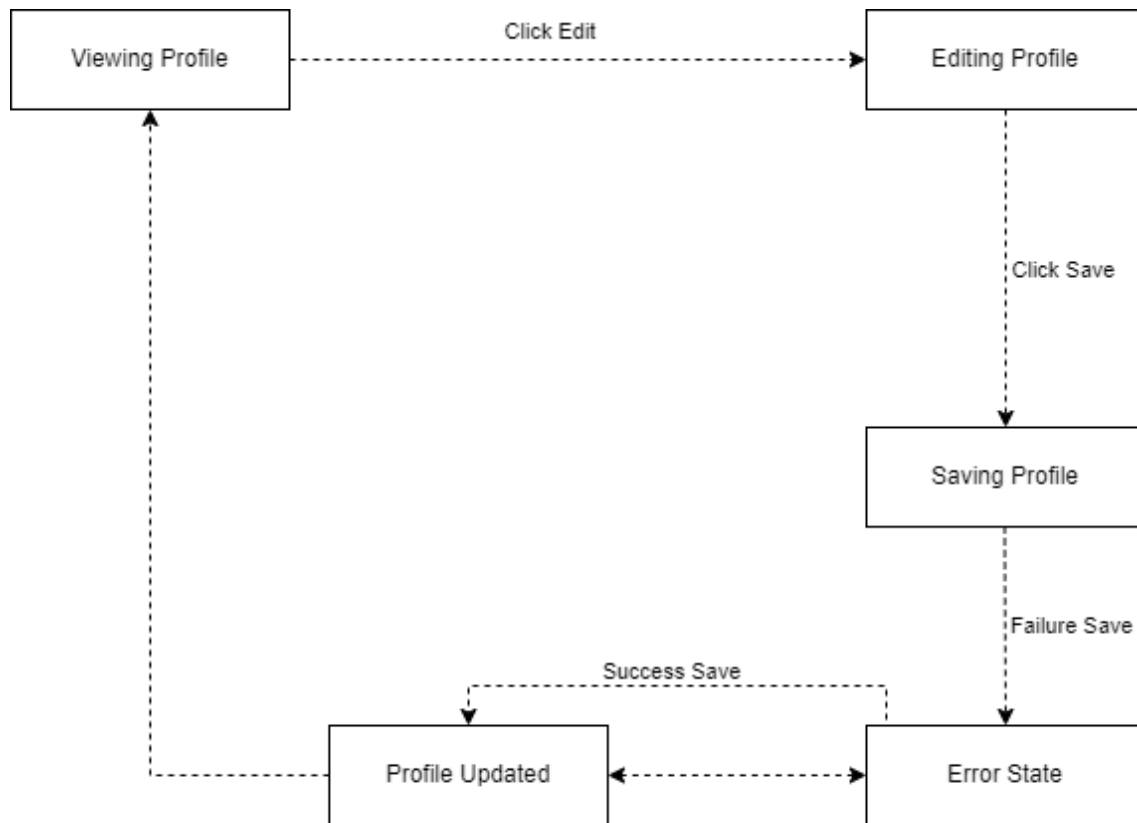


Figure 3.28: User Profile State Transition Diagram

2. Threads:

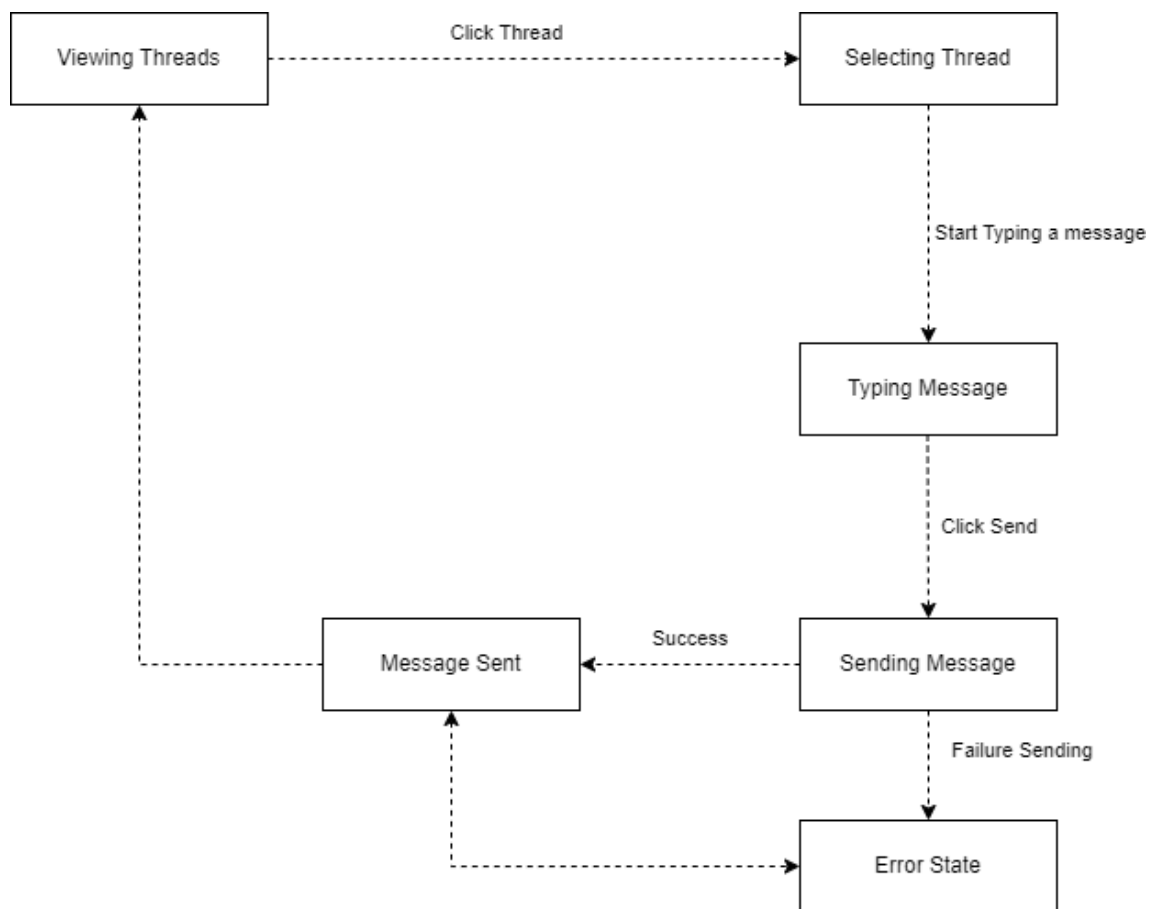


Figure 3.29: Threads State Transition Diagram

3. Synergy:

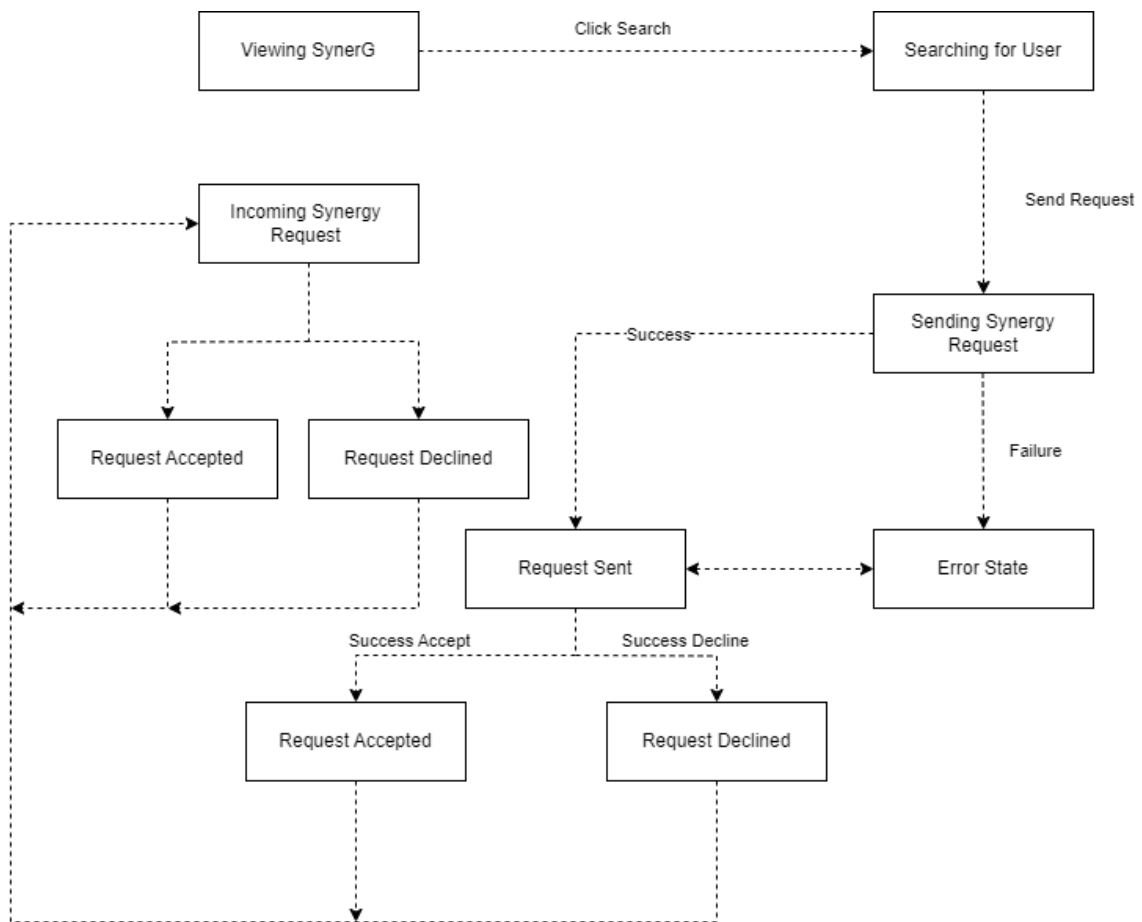


Figure 3.30: Synergy State Transition Diagram

4. Battlegrounds:

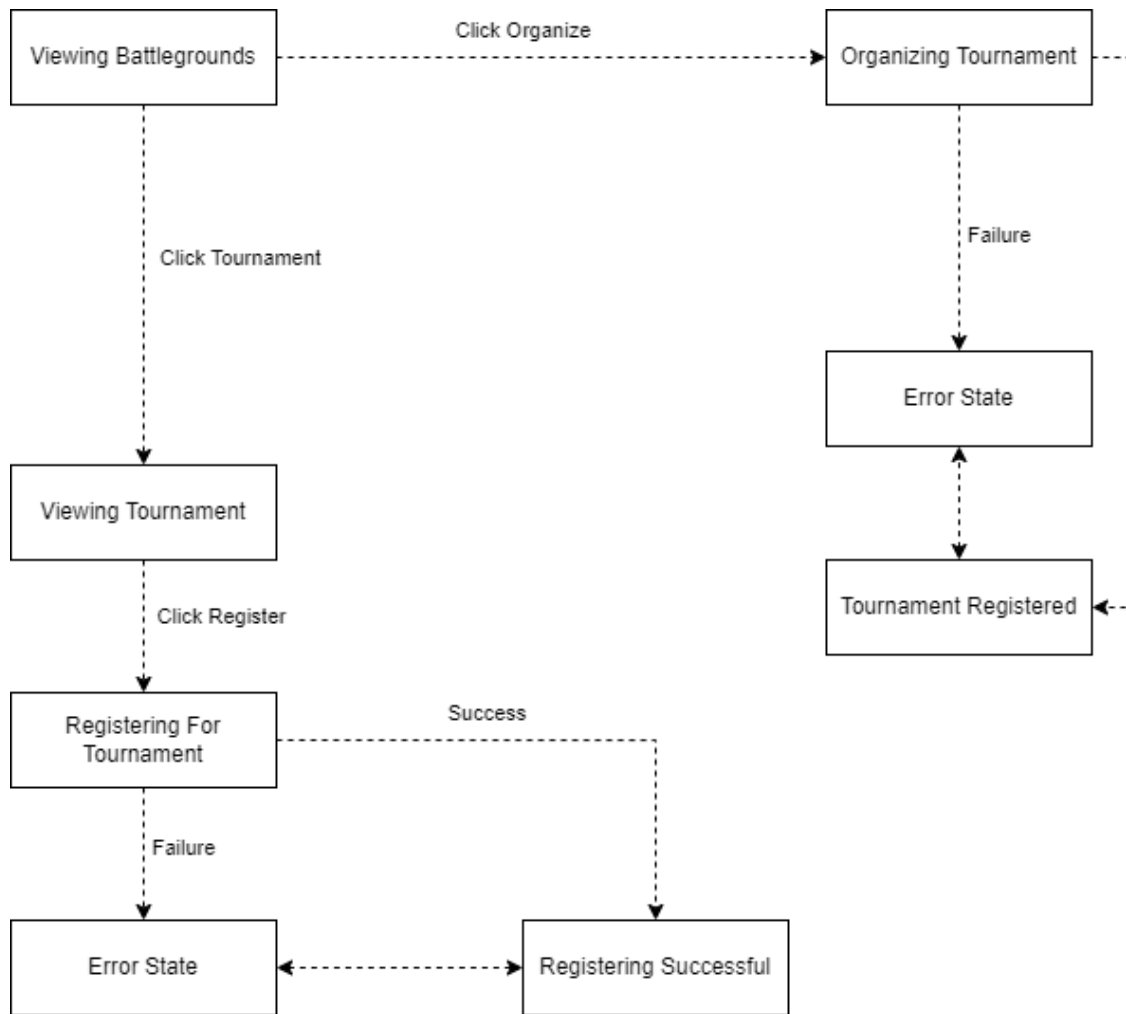


Figure 3.31: Battlegrounds State Transition Diagram

3.3 Data Design

ArenaX is a networking application with a lot of user and data-intensive features, its data design structure is such that it supports and ensures efficient data processing, storage and retrieval. The information domain of the app contains a number of entities including users, games, analytics, organization, connections, threads, content, battlegrounds and recruitment. These entities are processed and stored in the PostgreSQL. However Firebase is used in the Signup/Signin module for basic user data storage and authentication purposes. Below depicted is the ERD diagram of ArenaX's entities and their transformation into organized data structures.

3.3.1 Data Structures

Below are the entities/ utilized in the app.

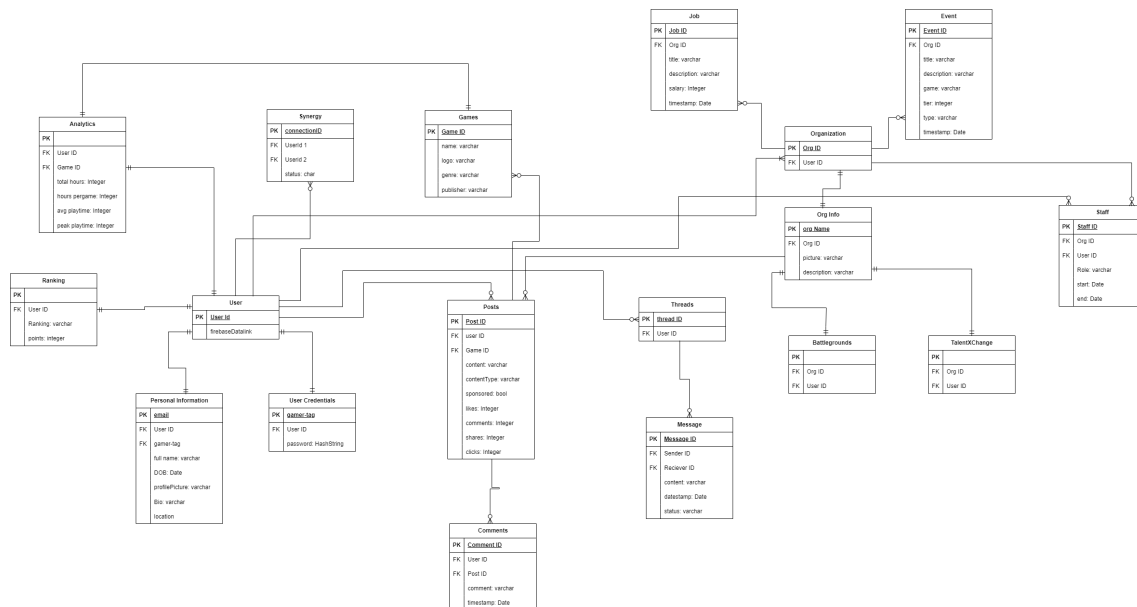


Figure 3.32: ER Diagram ArenaX

1. User

- **Attributes:** User ID (PK), link to user Data stored in Firebase.
- **Description:** Stores User IDs and serves as connection between Firebase and PostgreSQL.

2. User Credentials

- **Attributes:** gamer-tag (PK), User ID (FK), password,
- **Description:** This table stores users' account credentials .

3. Personal Information

- **Attributes:** email (PK), User ID (FK), gamer-tag (FK), full name, Date of Birth, Profile Picture, Bio, location.
- **Description:** This table stores user's personal information. This table stores the Urls of profile pictures stored in the Firebase storage.

4. Ranking

- **Attributes:** User ID (FK), ranking, total points.
- **Description:** This table stores the total points and ranking of an account based on its total hours spend in-game.

5. Games

- **Attributes:** Game ID (PK), name, logo, genre, publisher.
- **Description:** This table contains a games dictionary including all games and their relevant details.

6. Analytics

- **Attributes:** User ID (FK), Game ID (FK), total hours, hours per game, peak playtime, average playtime.
- **Description:** This table stores user's game stats that are calculated based on user's game-time.

7. Synergy

- **Attributes:** Connection ID (PK), User ID 1 (FK), User ID 2 (FK), connection status.
- **Description:** This table stores connections of each user with other users on the app.

8. Posts

- **Attributes:** Post ID (PK), User ID (FK), Game ID (FK) May be NULL, content, content type, timestamp, sponsored, likes, comments, shares, clicks.
- **Description:** This table stores details to a content posted by a user on the app in addition to storing the URL of the content stored in the Firebase storage.

9. Comments

- **Attributes:** Comment ID (PK), User ID (FK), Post ID (FK), comment, timestamp.
- **Description:** This table stores the comments posted by users on each post.

10. Threads

- **Attributes:** Thread ID (PK), User ID (FK)
- **Description:** Stores the chats of users.

11. Messages

- **Attributes:** Message ID (PK), Sender (FK), Reciever (FK), content, timestamp, message status
- **Description:** This table stores the data and its details sent over a message between two users.

12. Organization

- **Attributes:** Organization ID (PK), User ID (FK)
- **Description:** Stores the list of user and their registered organizations.

13. Organization Information

- **Attributes:** Org Name (PK), Org ID (FK), picture, description.
- **Description:** This table stores the information of a registered organization in addition to the url link of the organization's emblem.

14. Staff

- **Attributes:** Staff ID (PK), Org ID (FK), User ID (FK), role, start date, end date.
- **Description:** This table stores the data of each organization's staff including roles, and start and end date of a job.

15. Battlegrounds

- **Attributes:** Org ID (FK), User ID (FK).
- **Description:** Stores the interaction of organizations and users through the battlegrounds module.

16. Events

- **Attributes:** Event ID (PK), Org ID (FK), title, game, description, tier, type, timestamp.
- **Description:** This table stores information related to an event listed by an organization in the battlegrounds section.

17. TalentXchange

- **Attributes:** Org ID (FK), User ID (FK).
- **Description:** Stores the interaction of organizations and users through the TalentXchange module.

18. Jobs

- **Attributes:** Job ID (PK), Org ID (FK), title, description, salary, timestamp.
- **Description:** This table stores the information related to a job posted by an organization in the TalentXchange section.

3.3.2 Databases

The databases used for data storage in ArenaX include:

1. **Firebase** for basic user data storage including user credentials and personal information.
Details: As Firebase provides inbuilt security and authentication services, ArenaX utilizes these services of Firebase in the SignIn/SignUp module. In addition to this, Firebase storage is utilized to store all of that data on the app that exists in the form of pictures/videos.
2. **PostgreSQL** serves as the primary database for the app and stores all data present on the app including user Data, URLs to content and all other data entities within the system's boundary.
Details: PostgreSQL enables efficient data storage, processing and retrieval and so was selected as the primary database of the app.

Bibliography